



MAGNETICUM



Numerical Simulations of physical processes driving galaxy evolution

Lecture 1: Backbone Codes

Rhea-Silvia Remus

Canary Islands Winter School, 23.11.2021



Disclaimer

This lecture was designed for presentation with movie media. If no specific URL is provided, movies can be found at

www.usm.uni-muenchen.de/~rhea/teaching/movies

Movies are marked by a *

Introduction: This weeks topic?

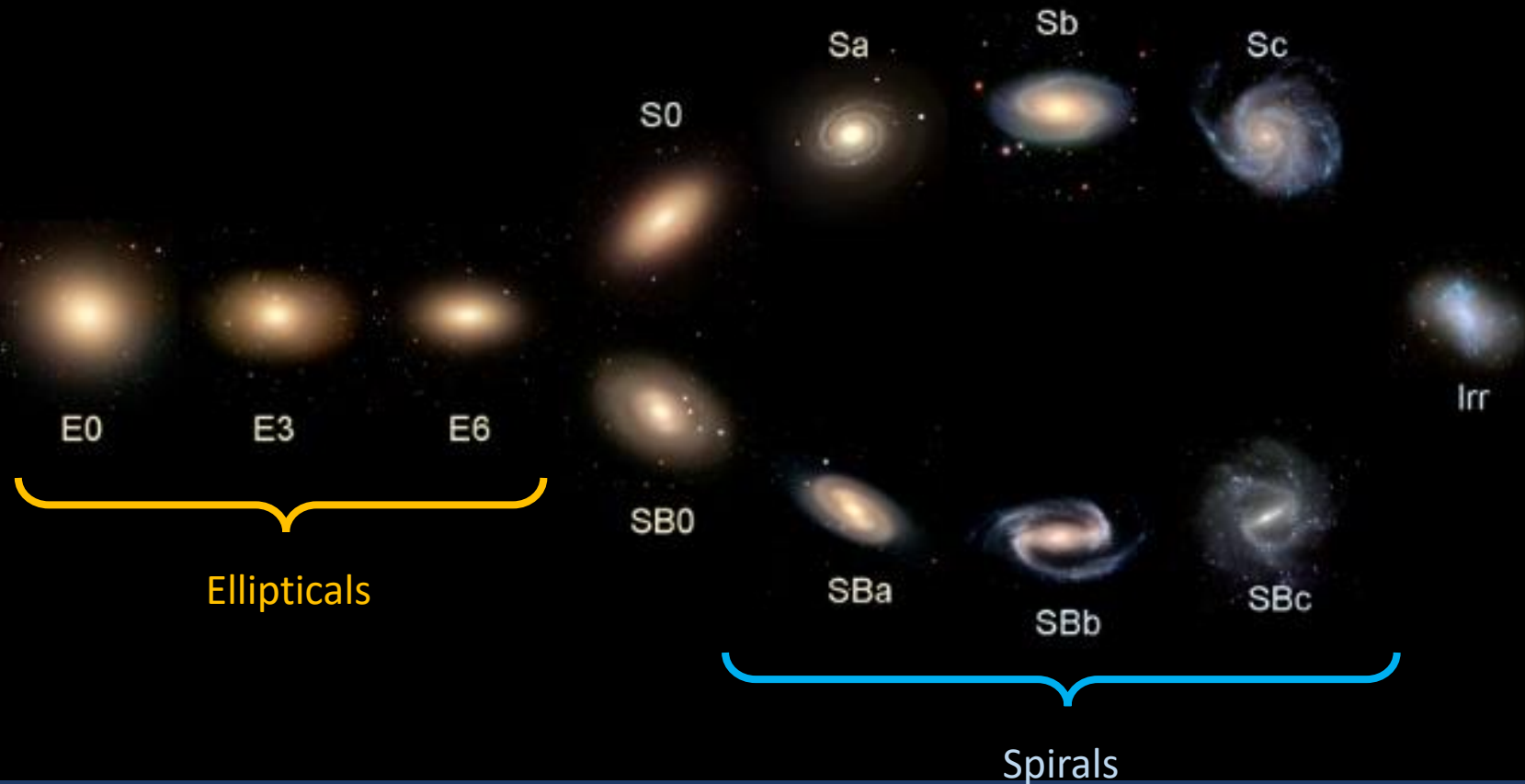
Numerical Simulations of physical processes driving galaxy evolution





Introduction: why do we care about galaxies?

Galaxies come in many different flavours, not just the well known regular shapes but a multitude of distorted features that need to be explained



NASA, ESA & The Hubble Heritage Team (STScI/AURA)



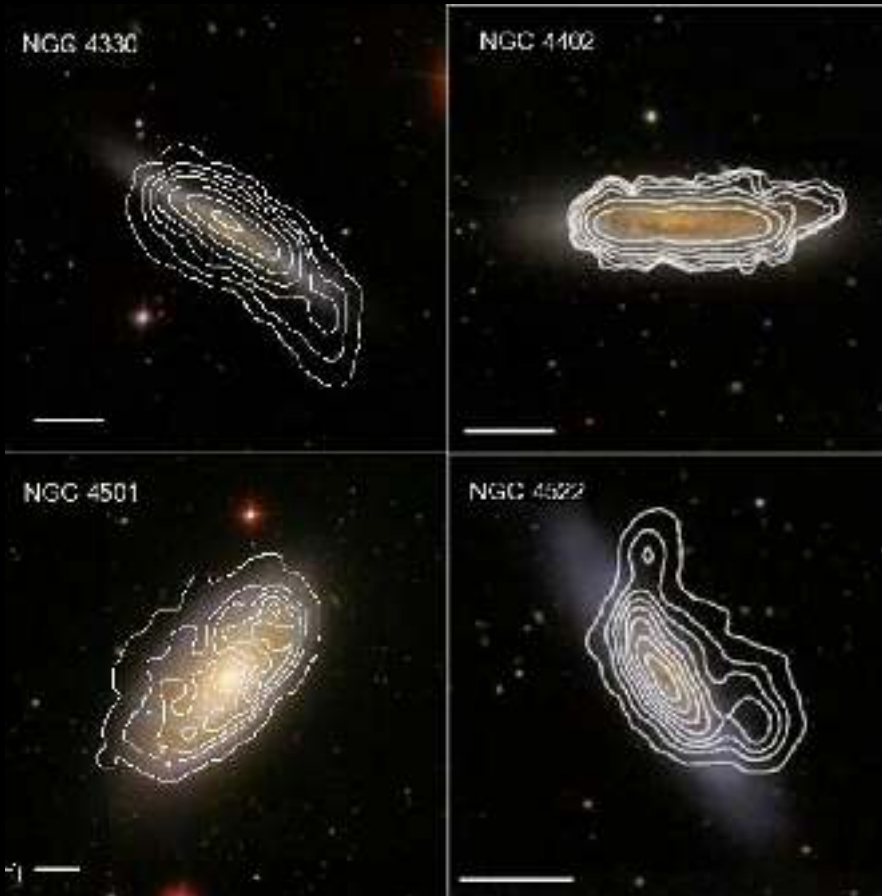
P.-A. Duc (CEA, CFHT) & Atlas3D



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Introduction: why do we care about galaxies?



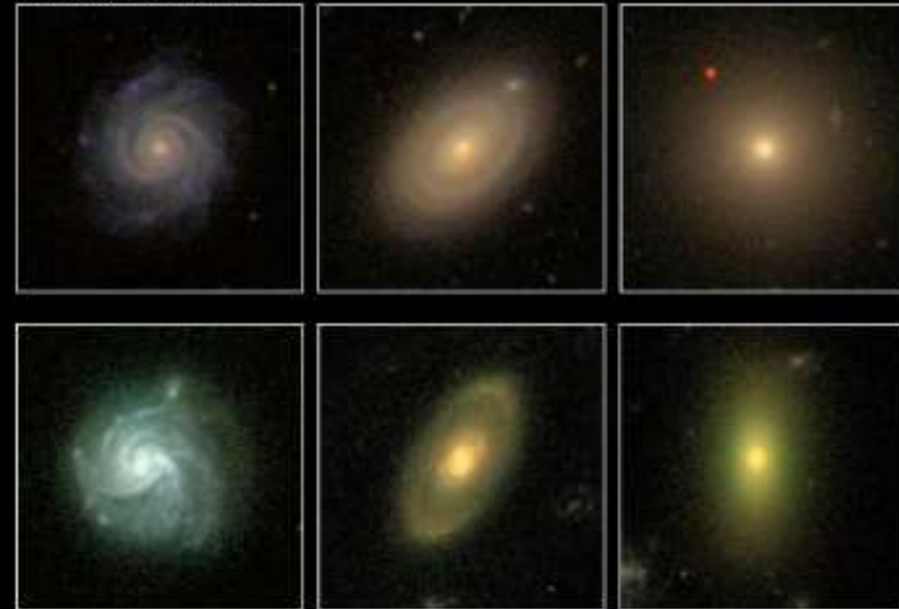
Wong et al., 2014



NASA/ESO

We also see environmental effects work on them, shown in the existence of for example jellyfish galaxies or red spiral galaxies

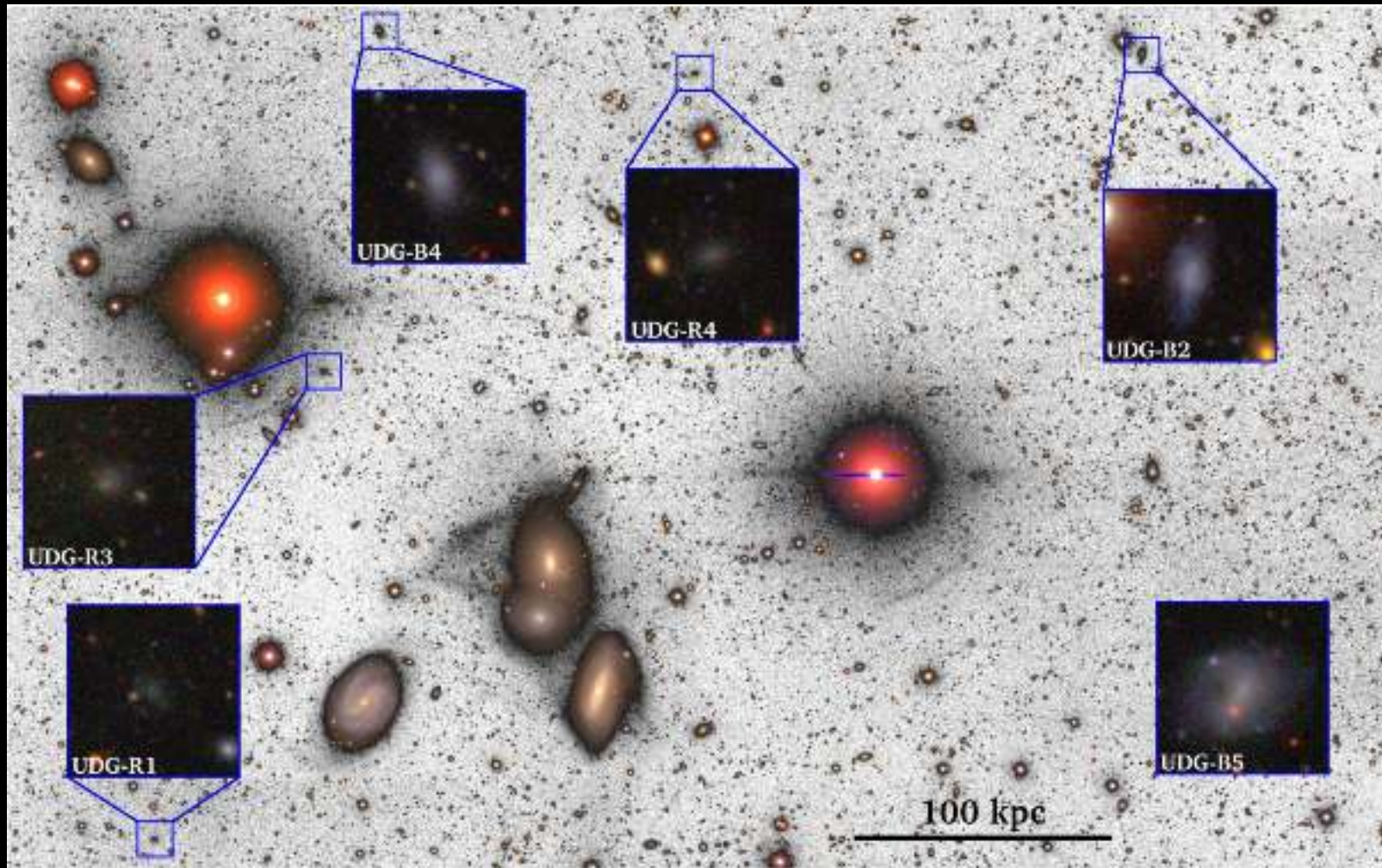
Galaxy Zoo : SDSS



STAGES : Hubble Space Telescope



Introduction: why do we care about galaxies?

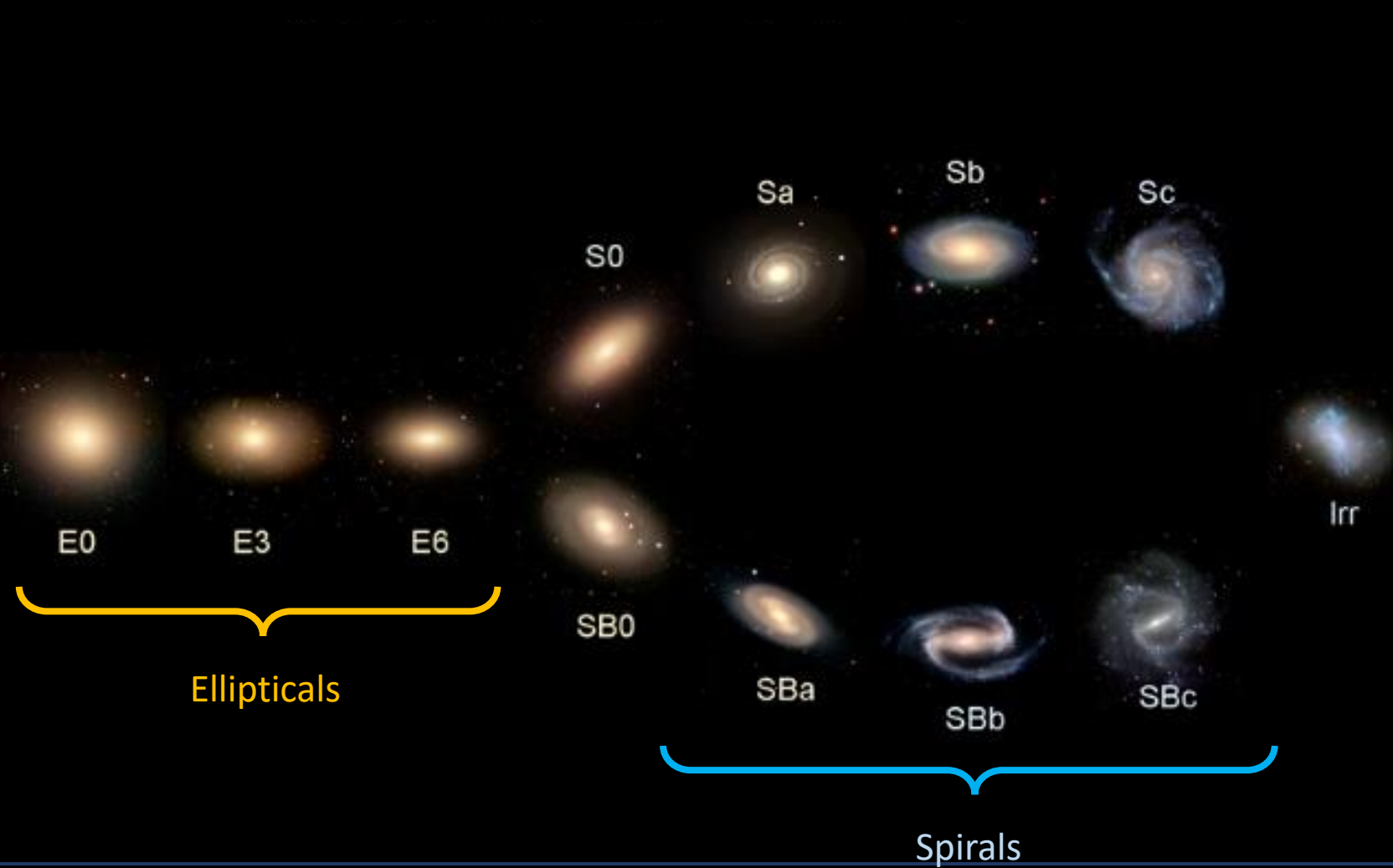


In addition, we find plenty of galaxies in the low-surface brightness regime, among which are the Ultra-Diffuse Galaxies (UDGs) which are especially numerous in the galaxy cluster environment

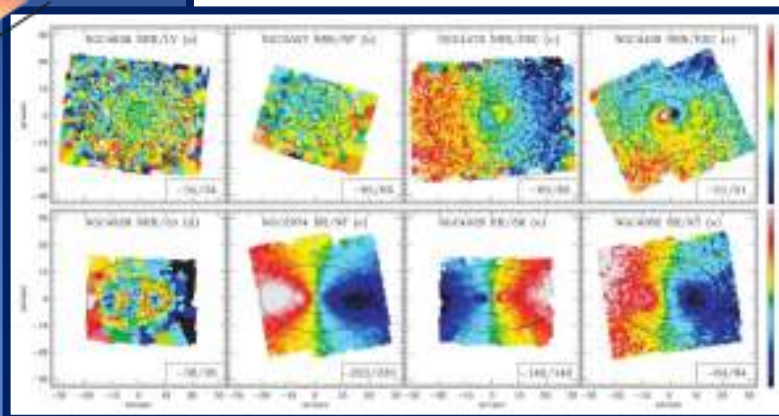
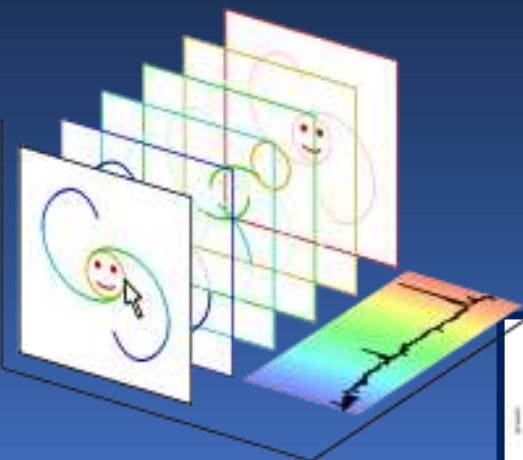
Roman & Trujillo 2017



Introduction: So how do we make all these different kind of galaxies?



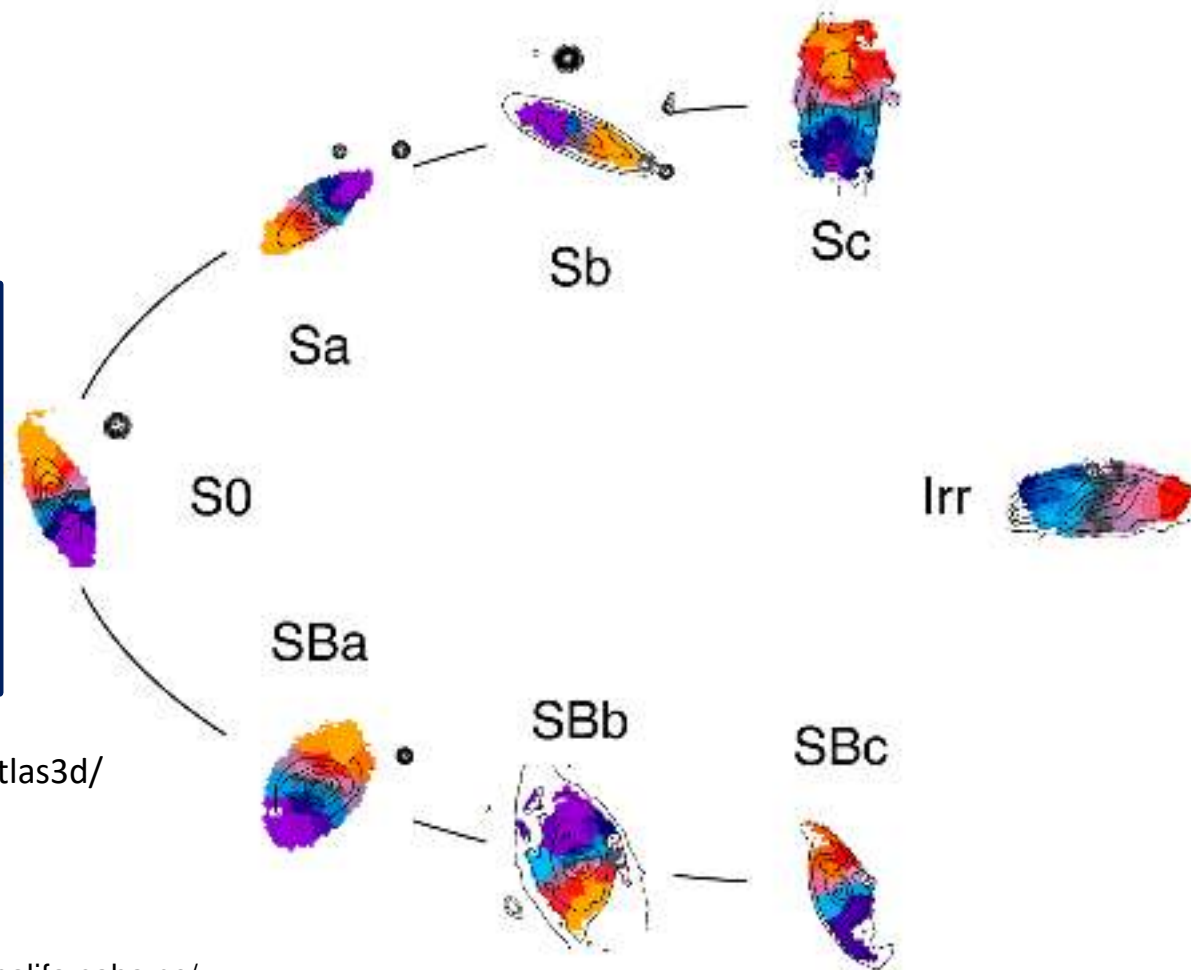
Introduction: why do we care about galaxies?



With the advent of Integral Field Spectroscopy, the features known to exist in galaxies were multiplied especially in the realm of quiescent galaxies

Atlas^{3D} Survey:
<https://www-astro.physics.ox.ac.uk/atlas3d/>

Pictures from the CALIFA survey: <http://califa.caha.es/>



Introduction: Numerical Simulations

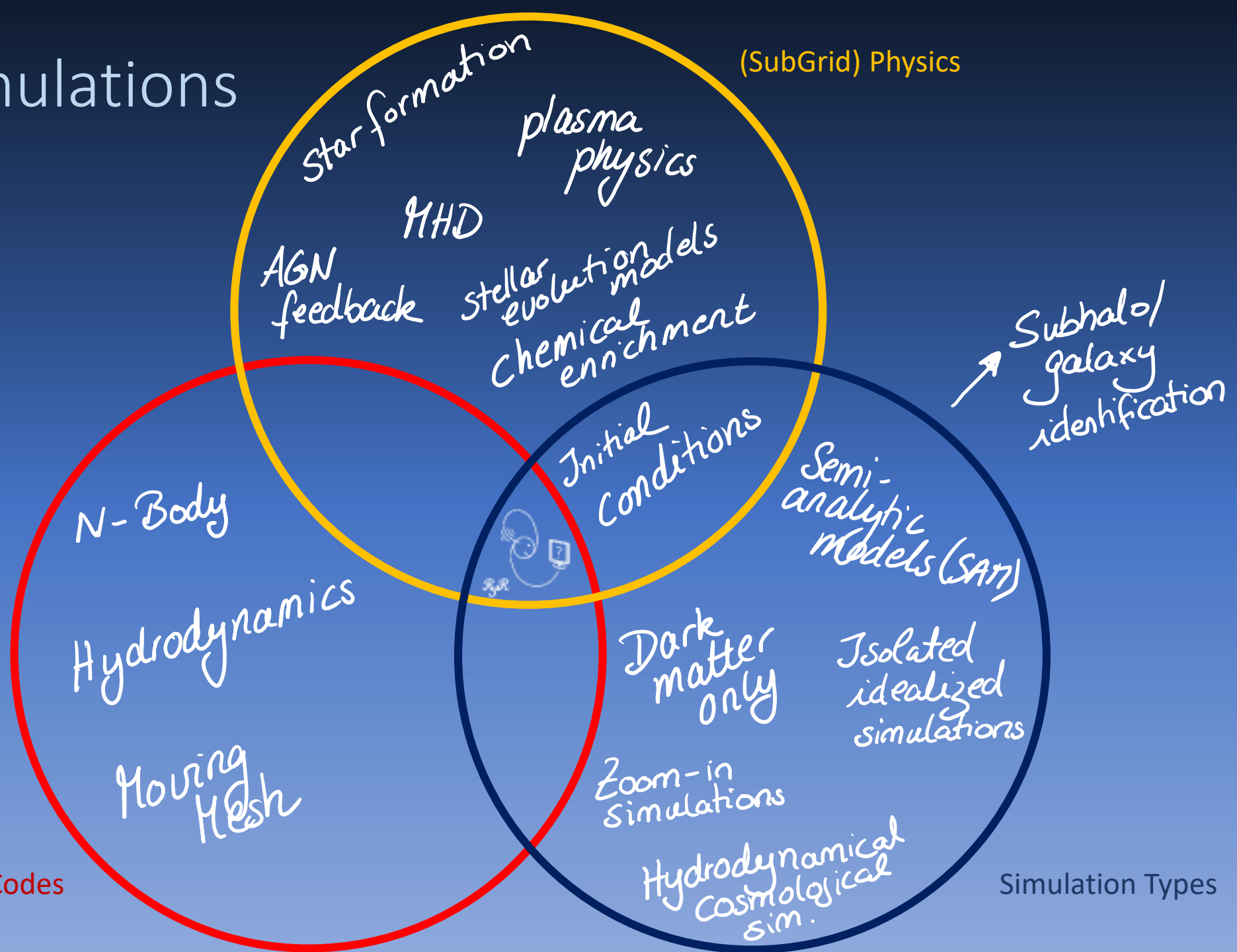
To understand the formation of galaxies, numerical simulations are an excellent tool to study the details of the processes that are responsible for the many faces of galaxies that can be observed. But how does that actually work?

Numerical Simulations

Backbone Codes

(SubGrid) Physics

Simulation Types





First Step: Gravity



N-Body Simulations

First Step: Gravity

Basic assumption: The matter in the Universe is **collision-less** and **non-relativistic** (Dark Matter is cold)



N-Body Simulations

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For **each particle i** : calculate gravitational force

$$F_{ij} = \frac{Gm_i m_j (r_j - r_i)}{\|r_j - r_i\|^3}$$



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So it requires N^2 calculations to calculate the equation of motion:

$$m_i \frac{d^2 r_i}{dt^2} = \sum_{\substack{j=1 \\ j \neq i}}^n \frac{Gm_i m_j (r_j - r_i)}{\|r_j - r_i\|^3}$$



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Integrate forces using Leap-Frog, Runge-Kutta, or higher order Integrators and obtain positions and velocities for all particles at every timestep



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Easy method, but
computationally expensive

Integrate forces using Leap-Frog, Runge-Kutta, or higher order Integrators and obtain positions and velocities for all particles at every timestep



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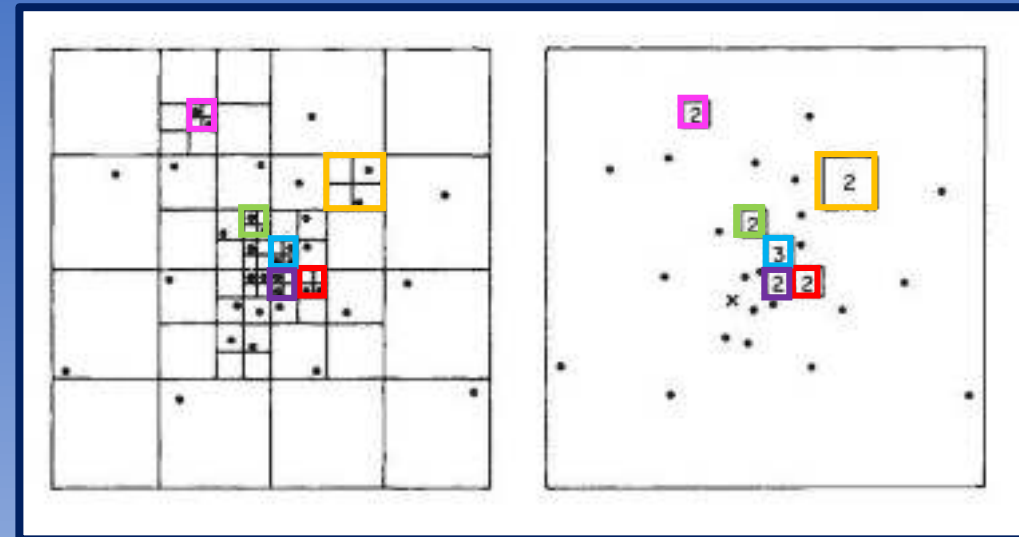
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Integrate forces using Leap-Frog, Runge-Kutta, or higher order Integrators and obtain positions and velocities for all particles at every timestep



Group together particles that are far away from particle i , and build a tree for the force calculation can limit the computational expense from N^2 to $N \log(N)$.

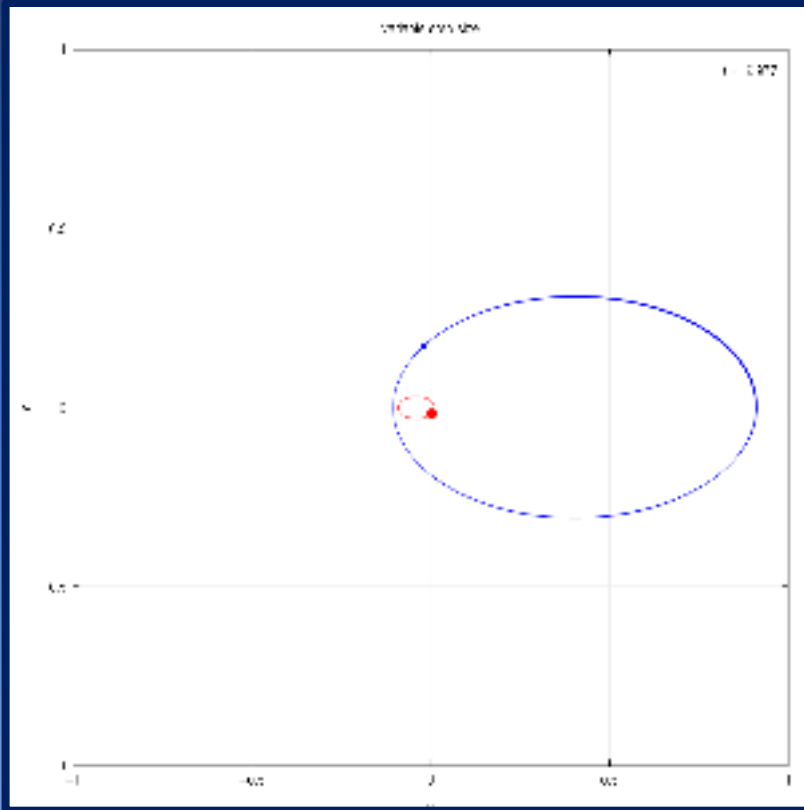




N-Body Simulations: Timestep Problem

*

Variable timestep



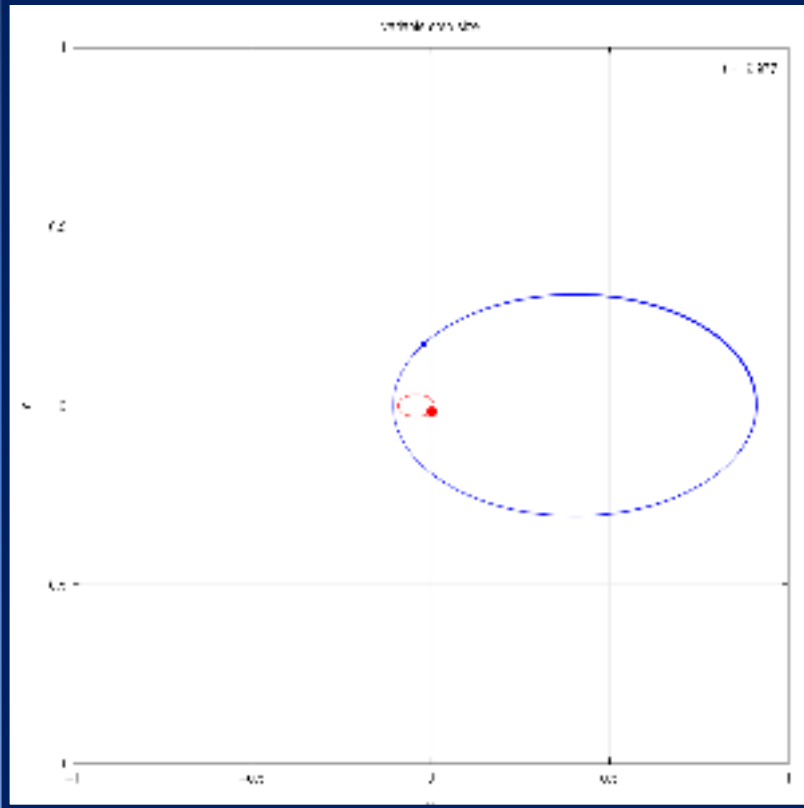
Credit: Tadziu Hoffmann



N-Body Simulations: Timestep Problem

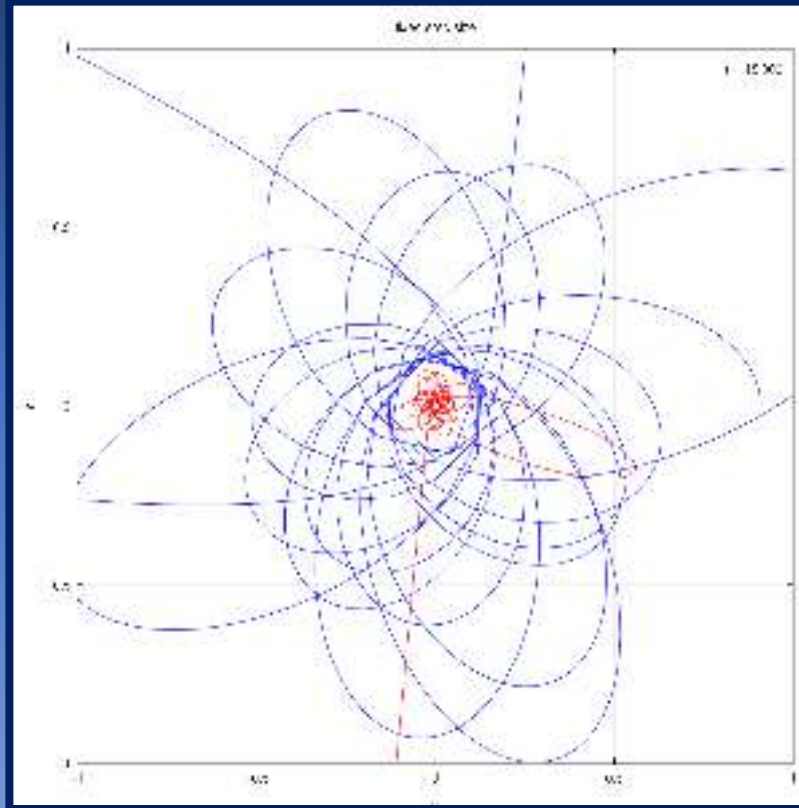
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Variable timestep



*

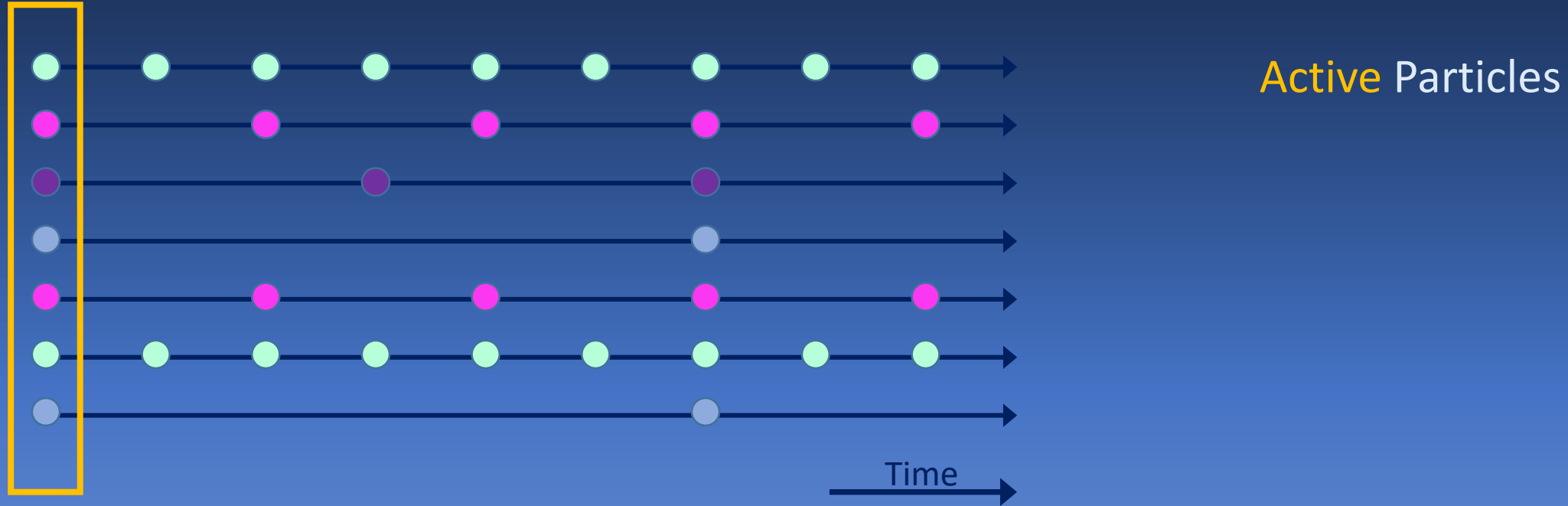
Fixed timestep



Credit: Tadeu Hoffmann

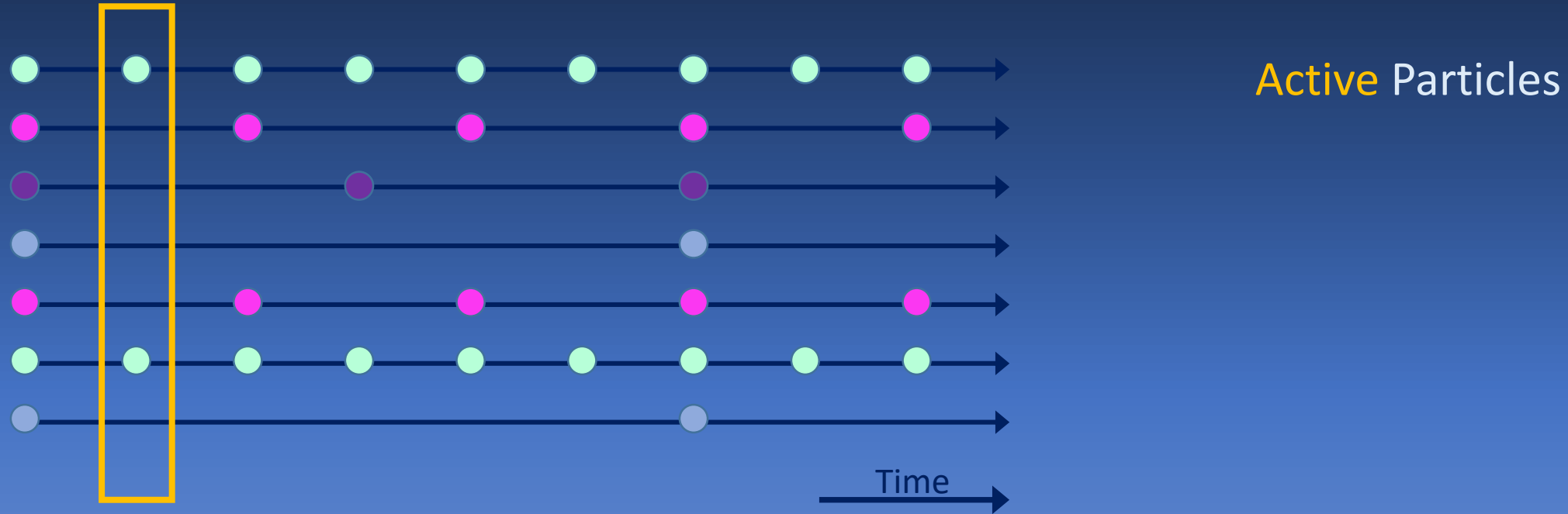


N-Body Simulations: Speedup I: Individual Timesteps



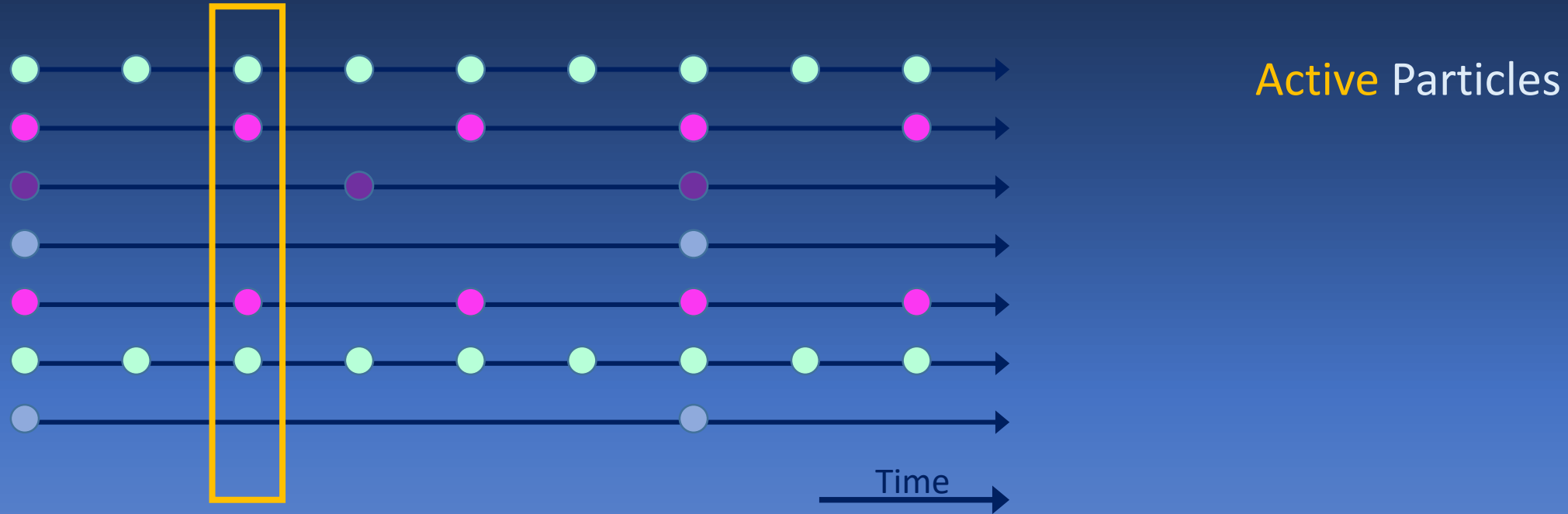
Using **individual timesteps** means that the Force needs to be calculated only for **active** particles

N-Body Simulations: Speedup I: Individual Timesteps



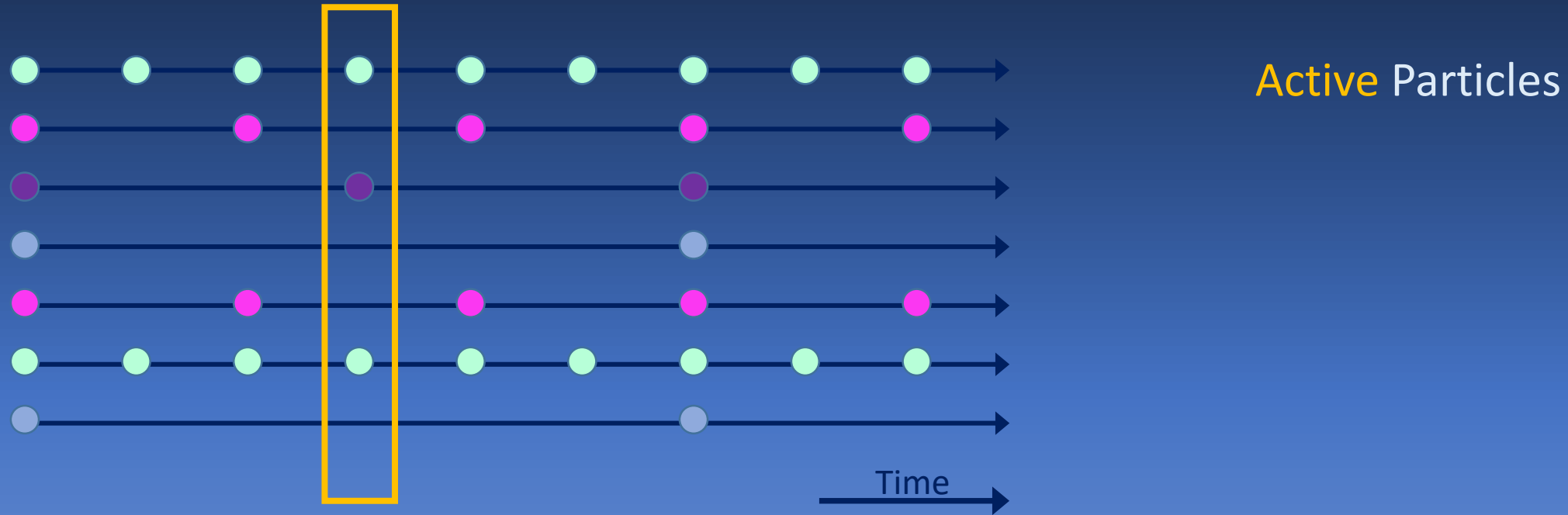
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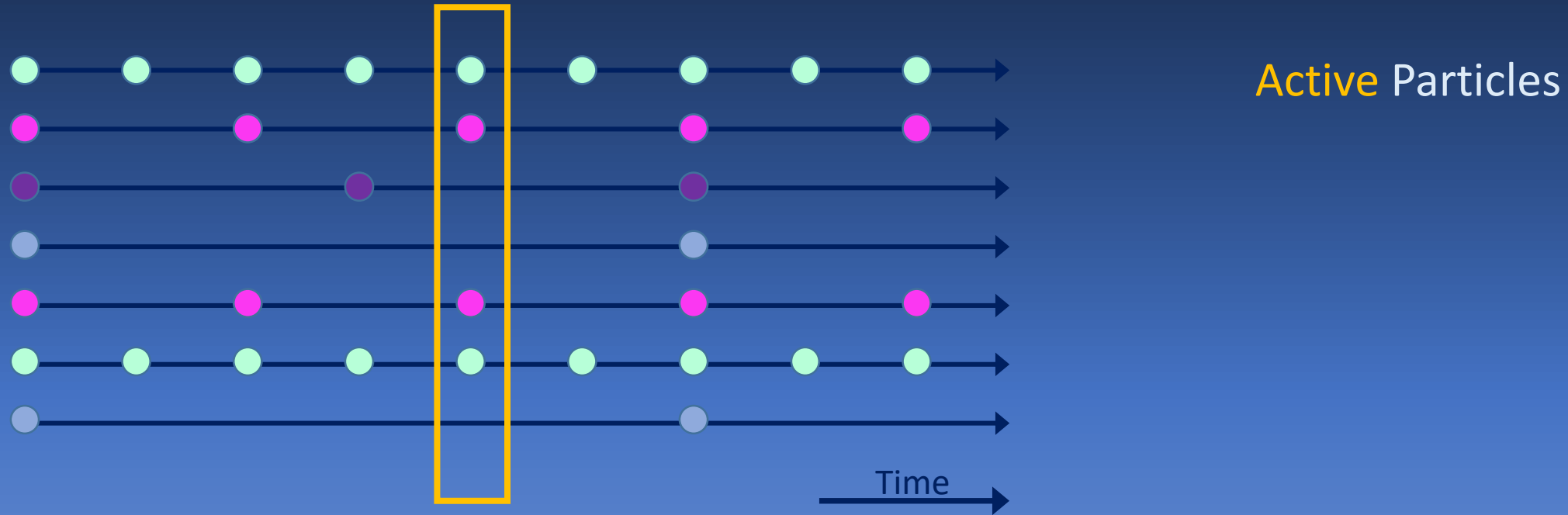
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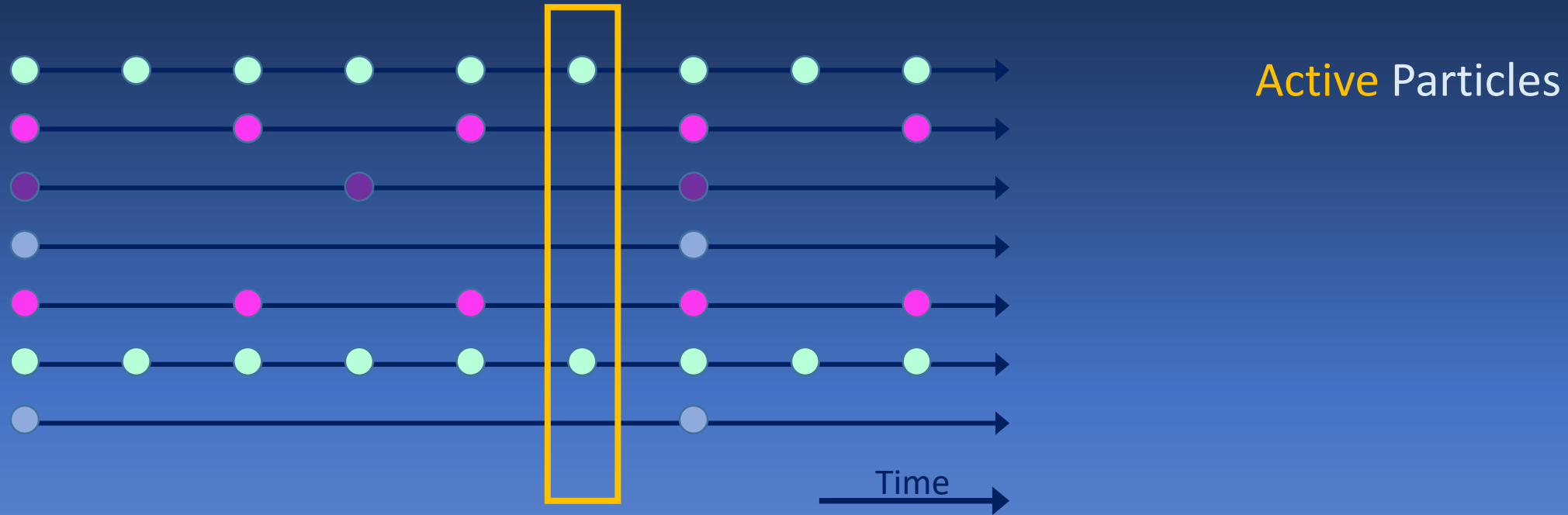
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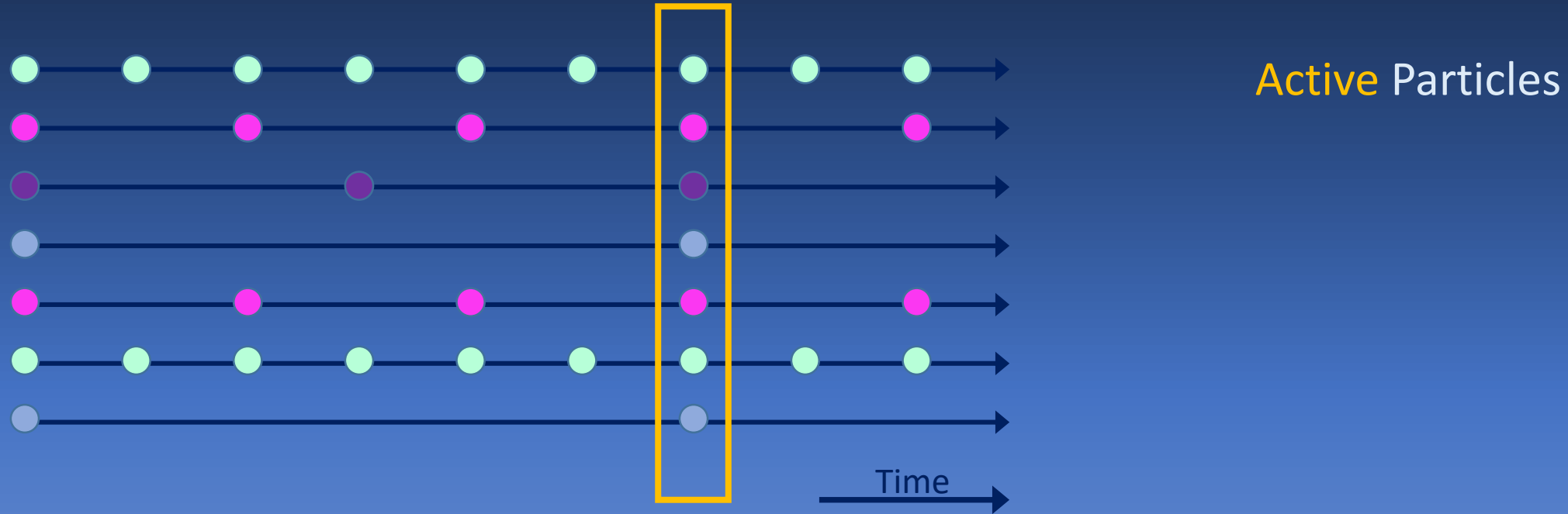
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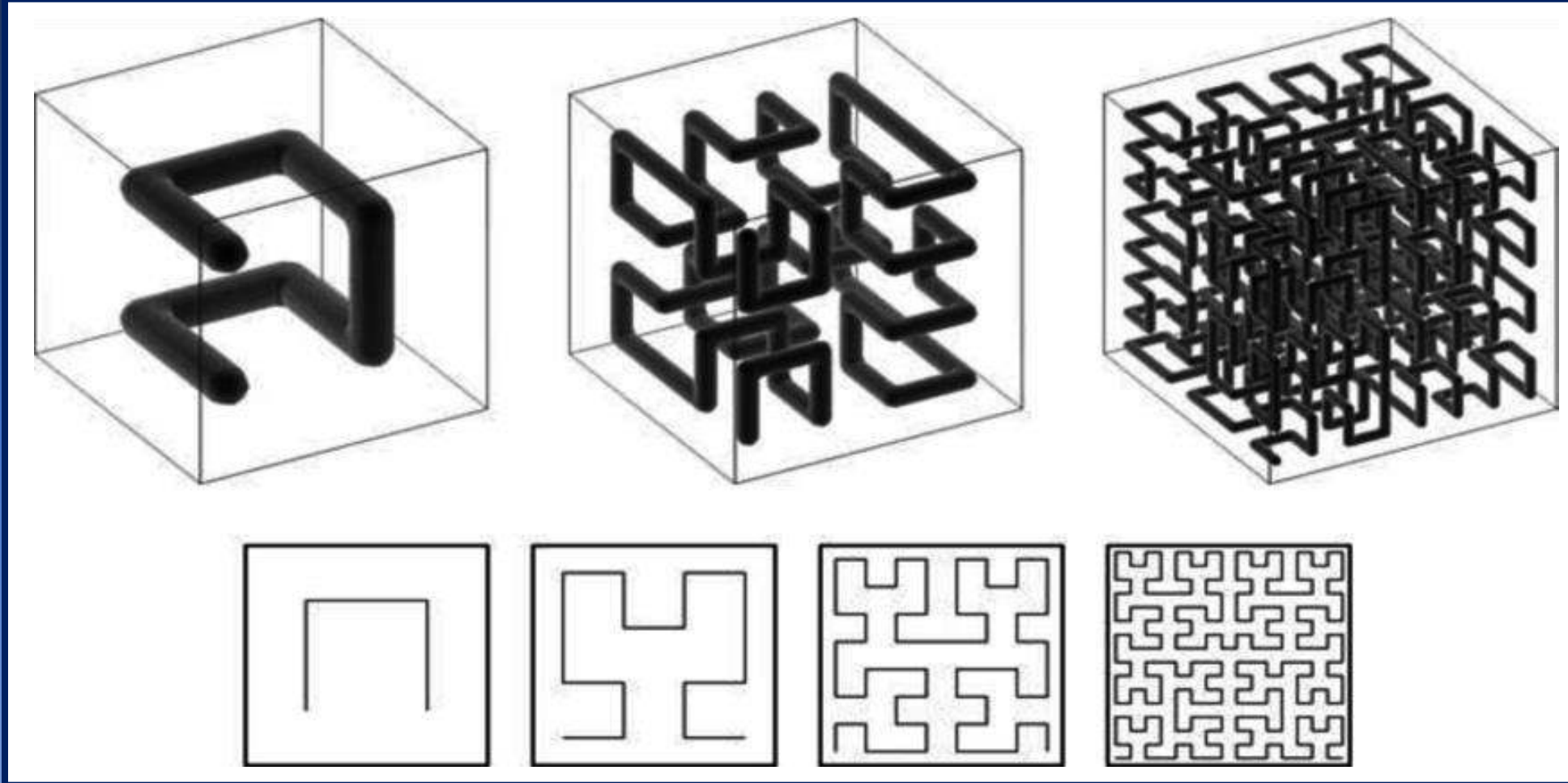
N-Body Simulations: Speedup I: Individual Timesteps



Using **individual timesteps** means that the Force needs to be calculated only for **active** particles



N-Body Simulations: Speedup II: Parallelize



Peano-Hilbert Curve

Distributing
particles
(memory) and
work load
(balance)
simultaneously



N-Body Simulations: Softening

Two Problems depending on the simulation setup, one solution:

- 1) In a `real particle simulation': If particles get very close, the **timesteps** are getting **extremely short**, causing numerical problems



N-Body Simulations: Softening

Two Problems depending on the simulation setup, one solution:

- 1) In a `real particle simulation': If particles get very close, the timesteps are getting extremely short, causing numerical problems
- 2) If the particles are actually not a representation of one object **but a group of objects**, they should not be able to collide.



N-Body Simulations: Softening

Two Problems depending on the simulation setup, one solution:

- 1) In a `real particle simulation': If particles get very close, the timesteps are getting extremely short, causing numerical problems
- 2) If the particles are actually not a representation of one object but a group of objects, they should not be able to collide.

To avoid this, we introduce an additional term into our force calculations, called **softening**:

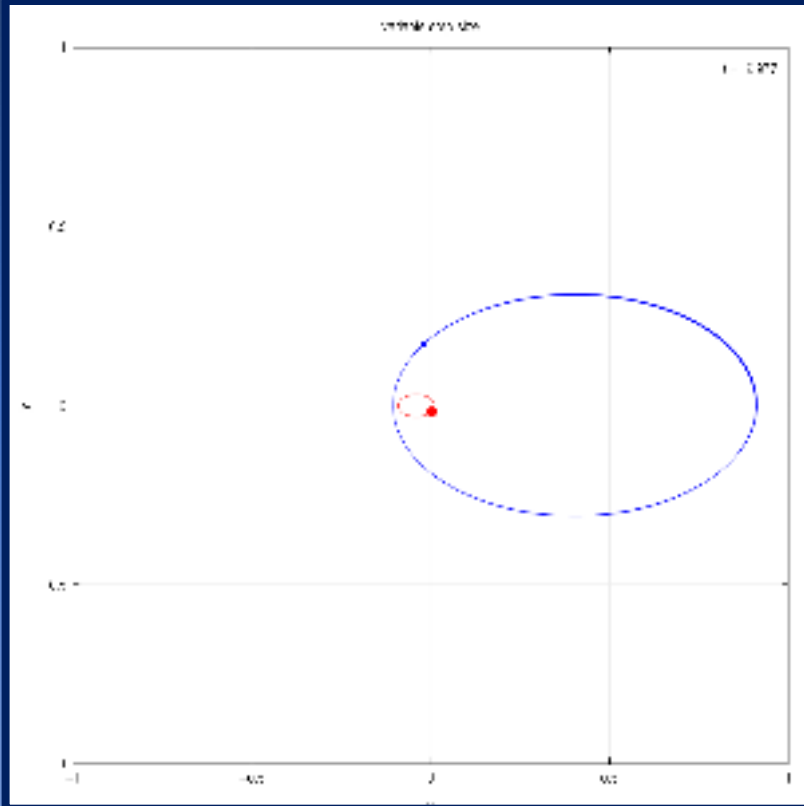
$$m_i \frac{d^2 r_i}{dt^2} = \sum_{\substack{j=1 \\ i \neq j}}^n \frac{G m_i m_j (r_j - r_i)}{\left(\|r_j - r_i\|^2 + \epsilon^2 \right)^{3/2}}$$



N-Body Simulations: Softening

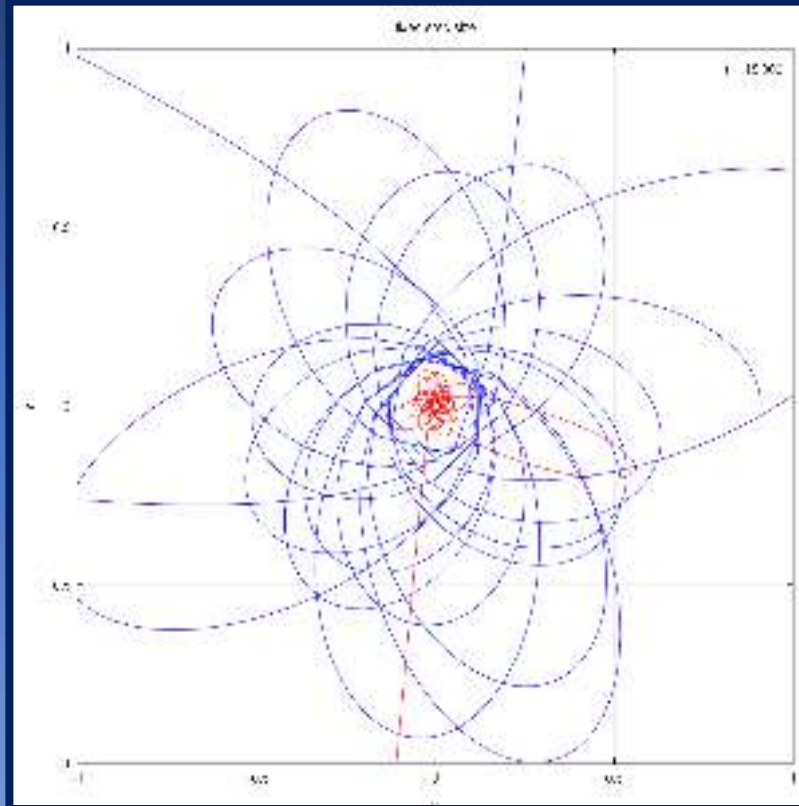
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Variable timestep



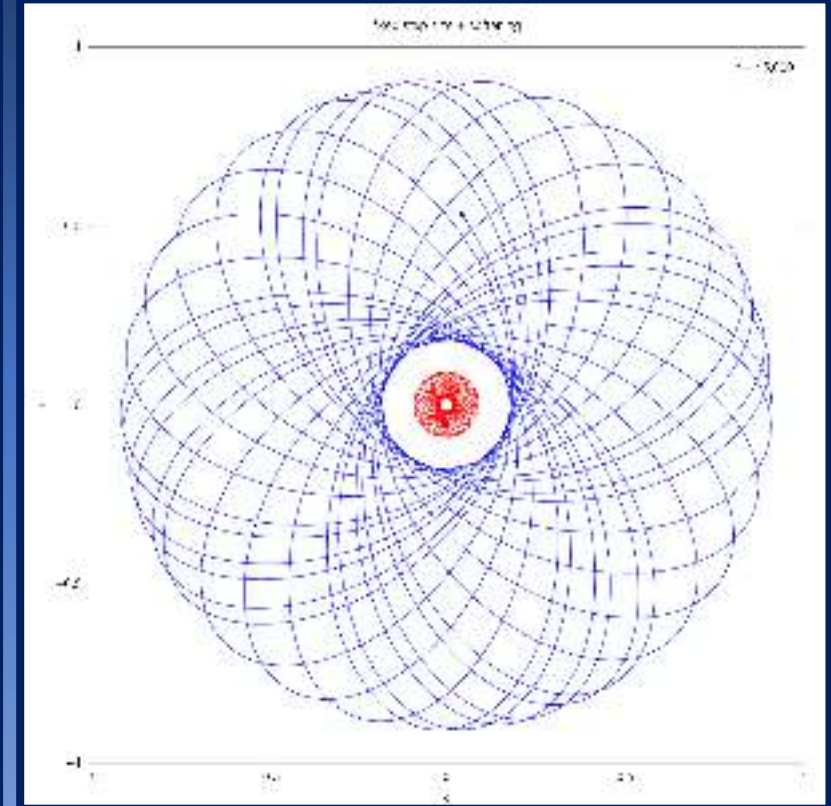
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Fixed timestep



*

With softening

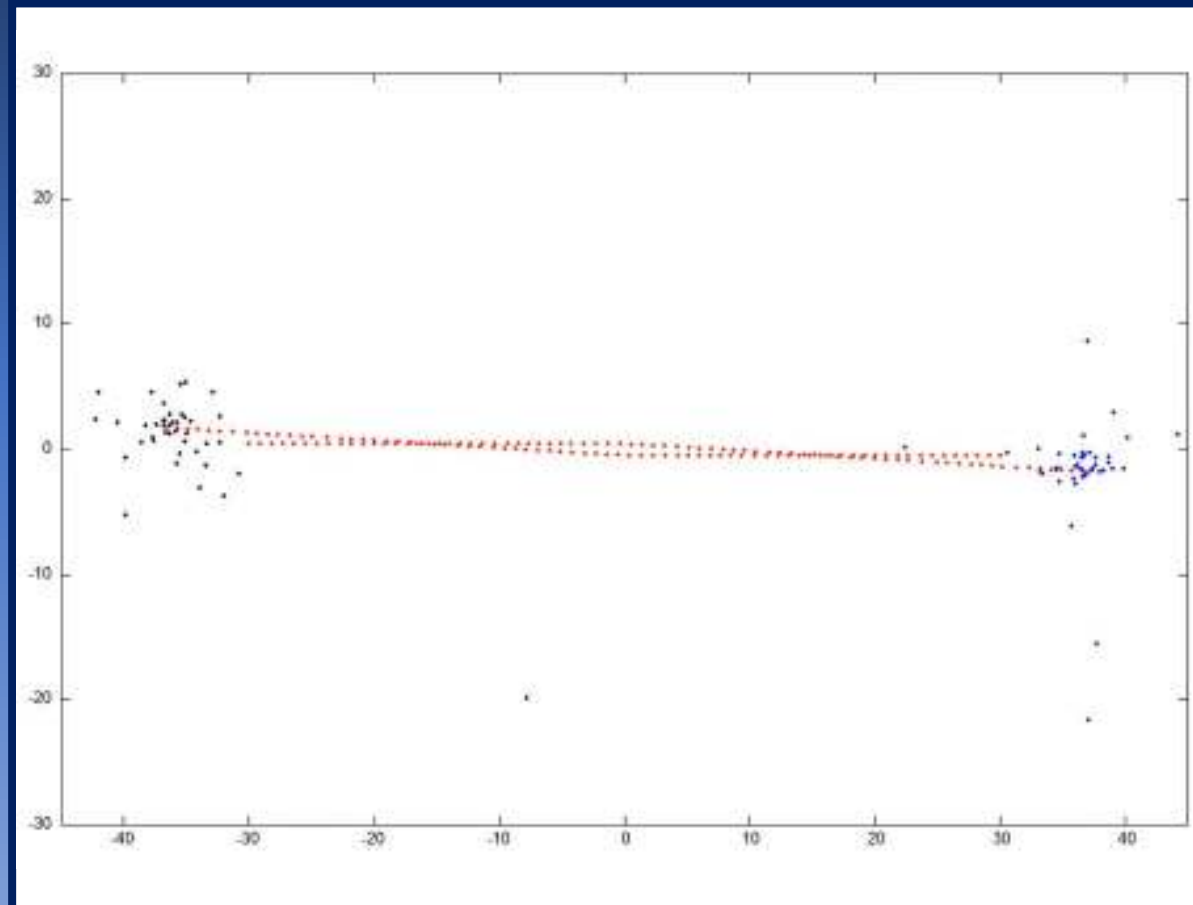


Credit: Tadeusz Hoffmann



Softening

Collision with Point Particles



*

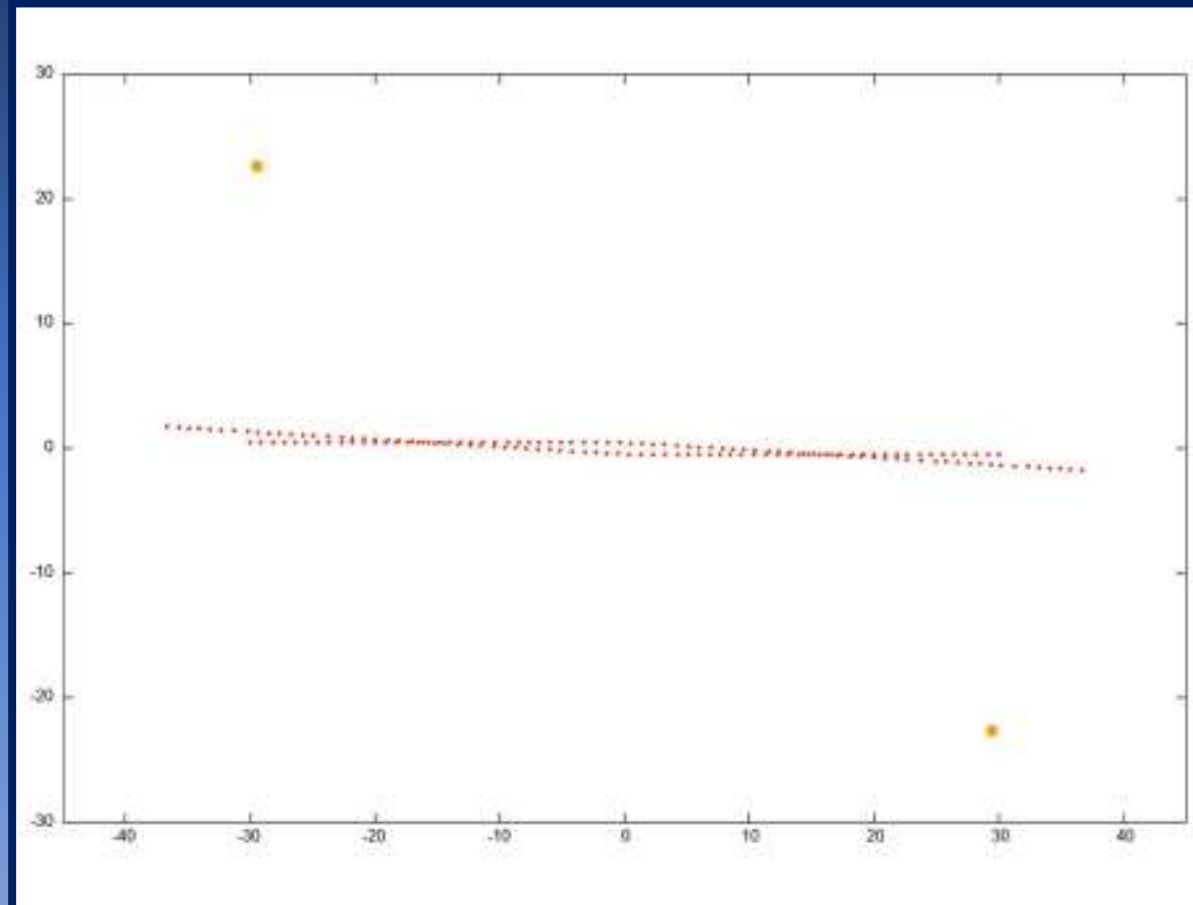
Red line: Movement of the centre of mass for both particle clouds

Credit: Lucas Valenzuela



Softening

Collision with Macro Particles, No softening



*

Red line: Movement of the centre of mass for original particle clouds

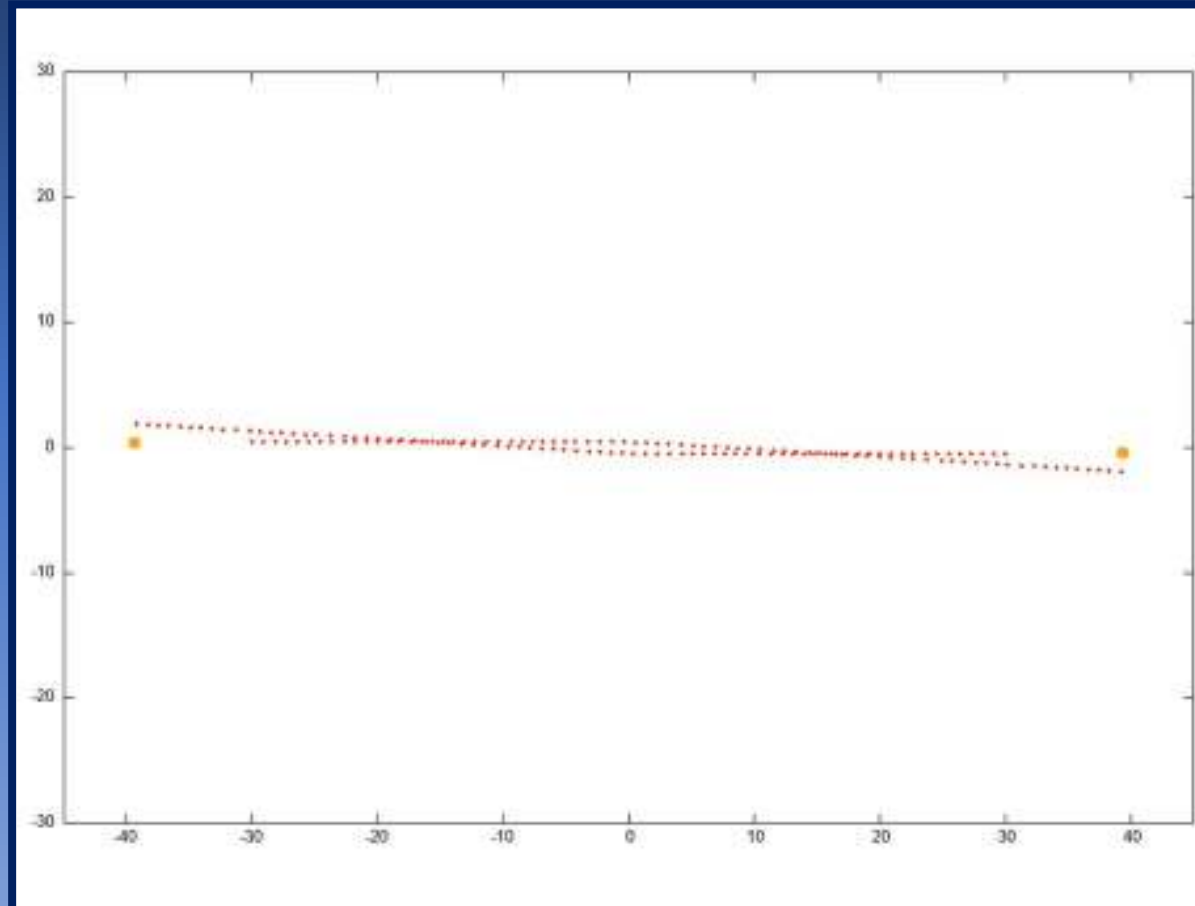
Yellow dots: New Macro Particles representing the point cloud

Credit: Lucas Valenzuela



Softening

Collision with Macro Particles, High softening ($\varepsilon = 5$)



*

Red line: Movement of the centre of mass for original particle clouds

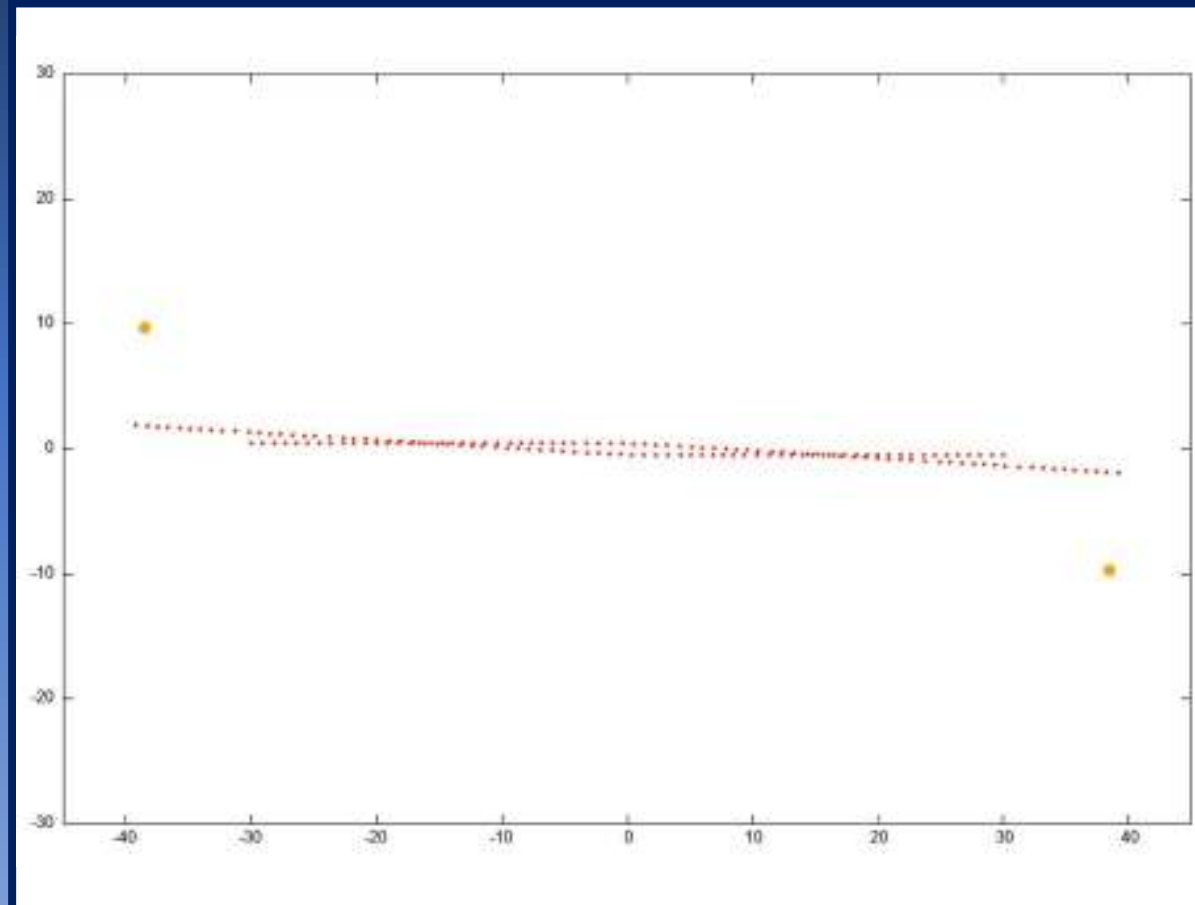
Yellow dots: New Macro Particles representing the point cloud

Credit: Lucas Valenzuela



Softening

Collision with Macro Particles, Low softening ($\varepsilon = 1$)



*

Red line: Movement of the centre of mass for original particle clouds

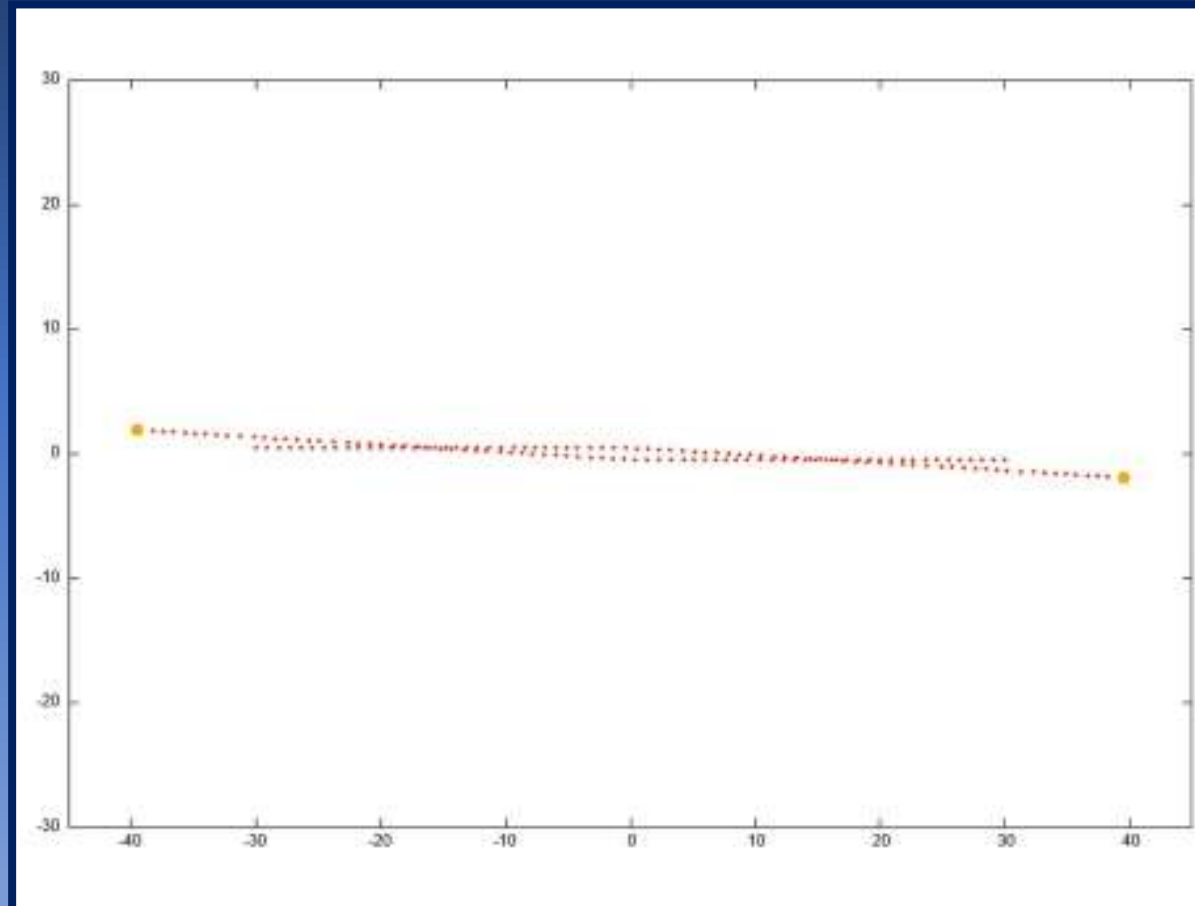
Yellow dots: New Macro Particles representing the point cloud

Credit: Lucas Valenzuela



Softening

Collision with Macro Particles, perfect softening ($\varepsilon = 2.8$)



*

Red line: Movement of the centre of mass for original particle clouds

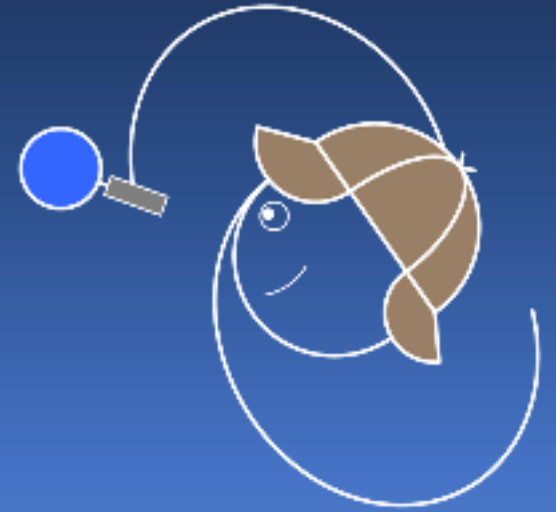
Yellow dots: New Macro Particles representing the point cloud

Credit: Lucas Valenzuela

Take a Break: Riddle!

*I am the reason you stir your coffee,
I am what keeps the plane in the air.
I caused Mark Antony to lose his last battle,
Leading to an epic love affair.
Without me
There would be
No light in the Universe.*

Who am I?





Problems with the Gas!



Problems with the Gas!

Including Hydrodynamics



Problems with the Gas!

While Stars and Dark Matter can be described to our current knowledge well as point particles, for gas this is more of an issue, as gas is a much more diffuse component where the **hydrodynamical equations** become important.



Problems with the Gas!

While Stars and Dark Matter can be described to our current knowledge well as point particles, for gas this is more of an issue, as gas is a much more diffuse component where the hydrodynamical equations become important.

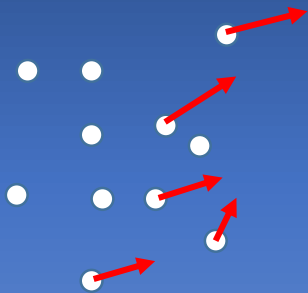
Two ways to treat gas:

- 1) discretize **mass**
- 2) discretize **space**



Problems with the Gas!

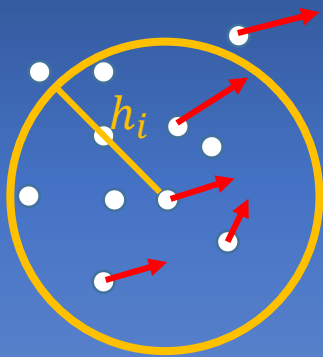
Treat gas as discretized particles:
Smoothed Particle Hydrodynamics
(Lagrangian method)





Problems with the Gas!

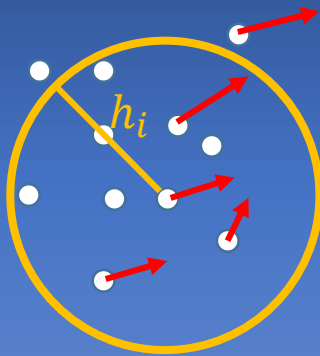
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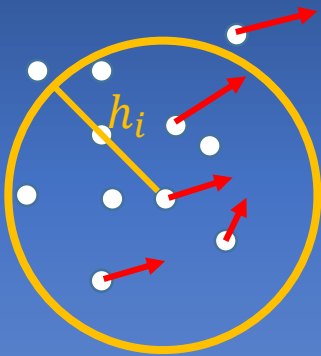


Discretizing mass means that the errors in the density filling results in **errors in the volume** measure. However, both the **equations and the discretization** are **Galilei-invariant**



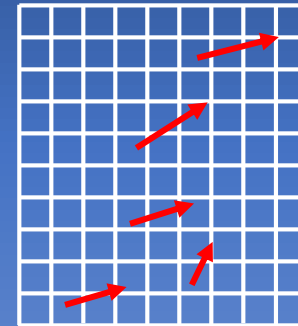
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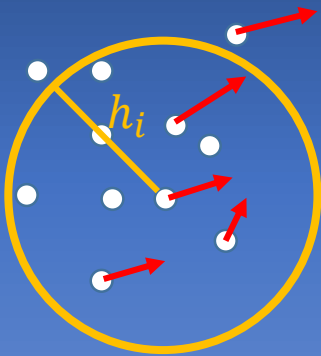
Discretize space: Treat gas within a
grid: Mesh Hydrodynamics
(Eulerian method)





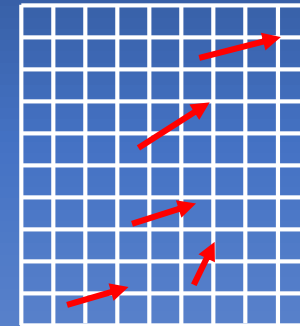
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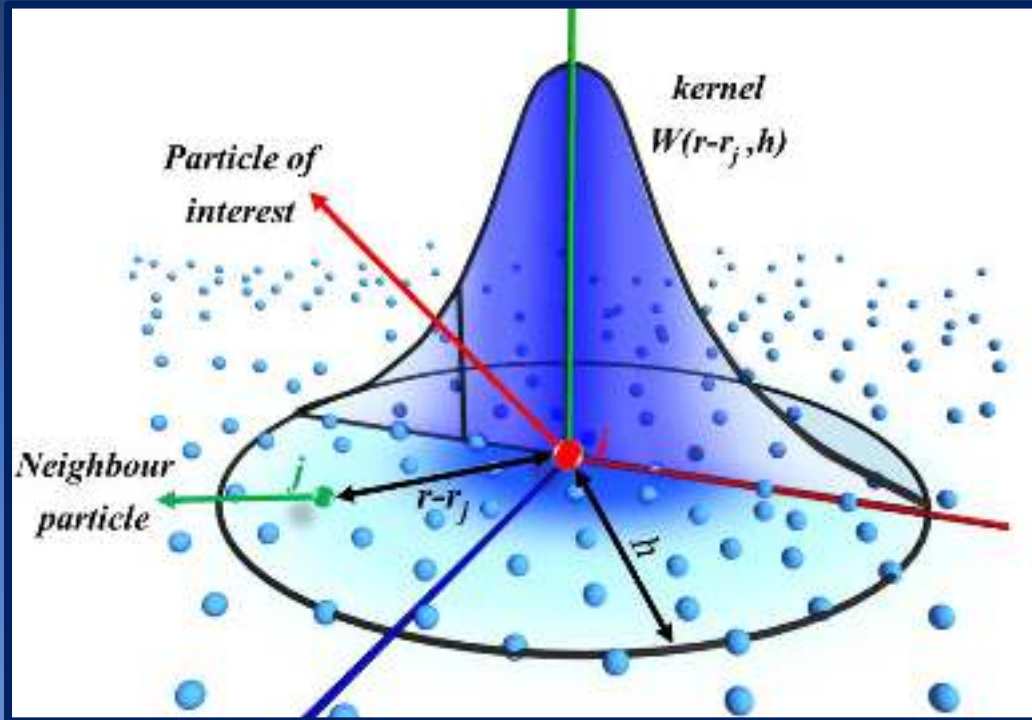


Discretizing space means that volume is fixed and errors in the density results in **mass loss of growth**. While the equations here are also Galilei-invariant, the **discretization is not!**



SPH

Smoothed Particle Hydrodynamics



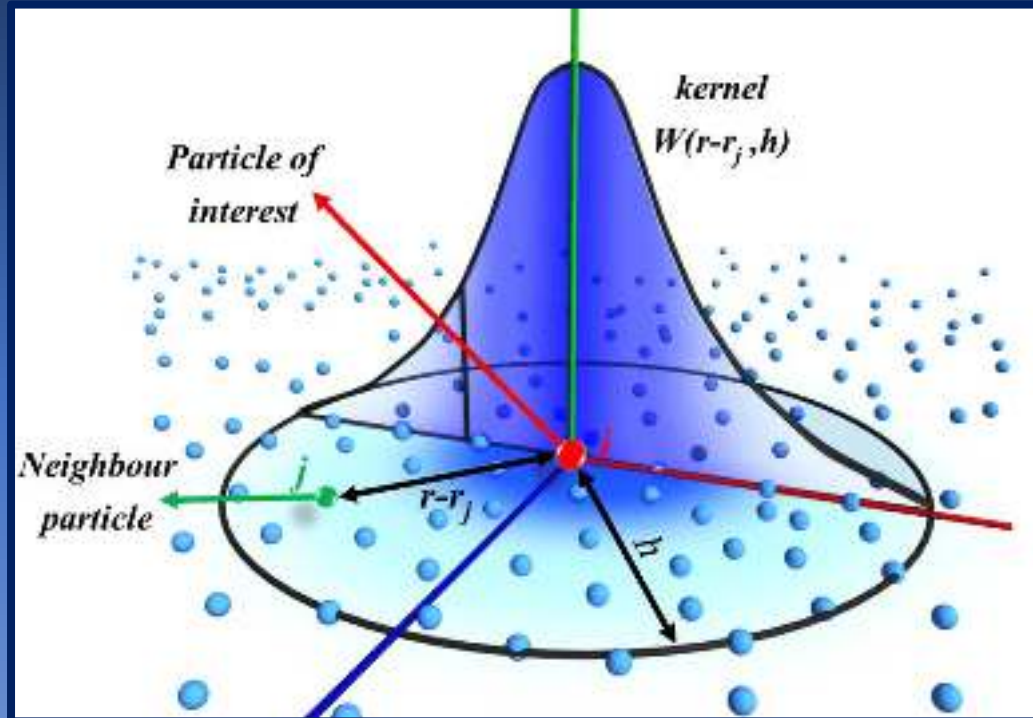
PhD Thesis Giuseppe Giorgi

$$\langle A(\mathbf{r}) \rangle = \int W(\mathbf{r} - \mathbf{r}', h) A(\mathbf{r}') d^3 r'$$
$$\langle A_i \rangle = \sum_{j=1}^N \frac{m_j}{\rho_j} A_j W(\mathbf{r}_{ij}; h_i)$$



AMR

Smoothed Particle Hydrodynamics

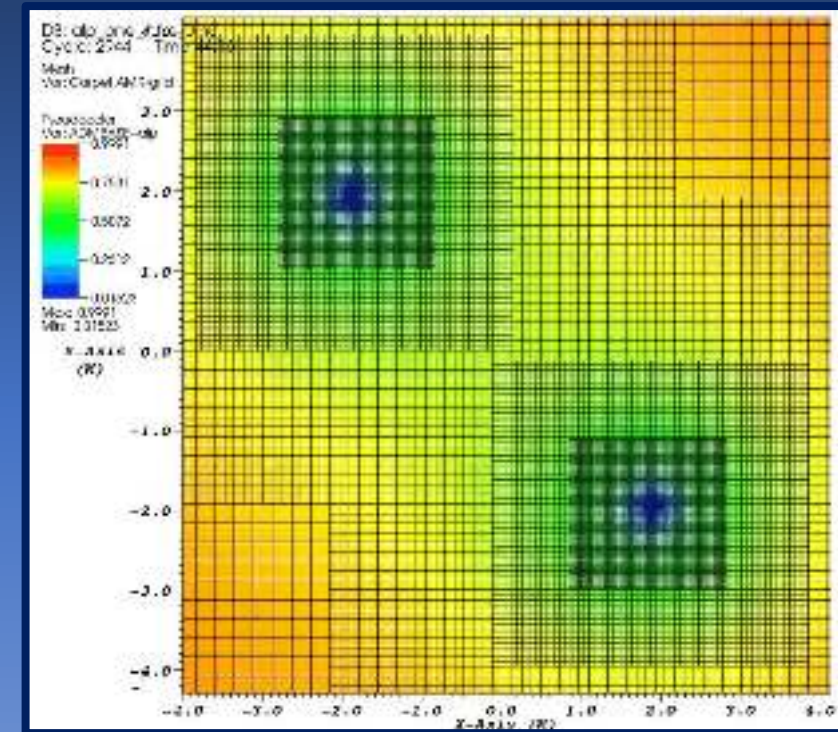


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Adaptive Mesh Refinement

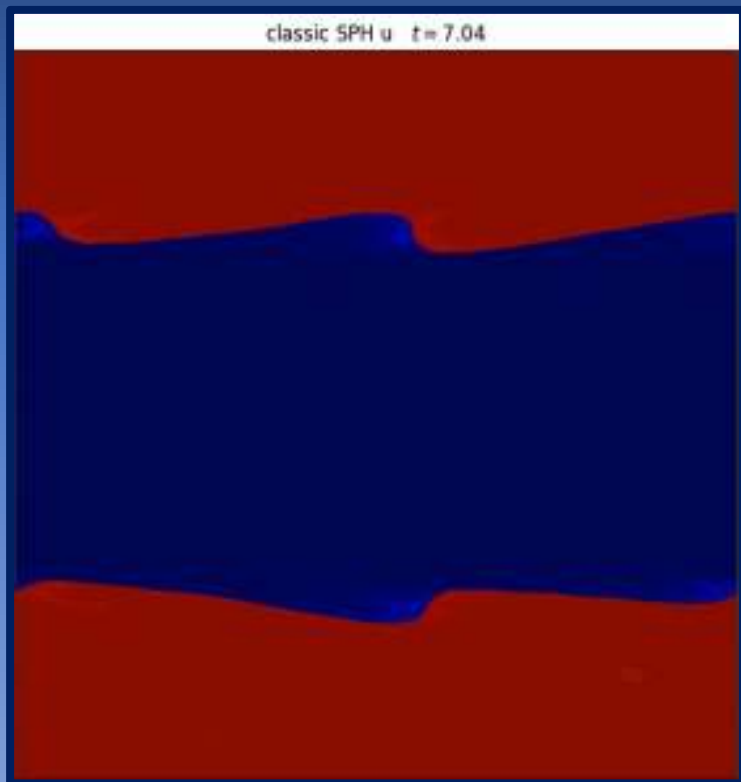


Janiuk & Charzynski 2016



SPH

Treat gas as discretized particles:
Smoothed Particle Hydrodynamics
(Lagrangian method)



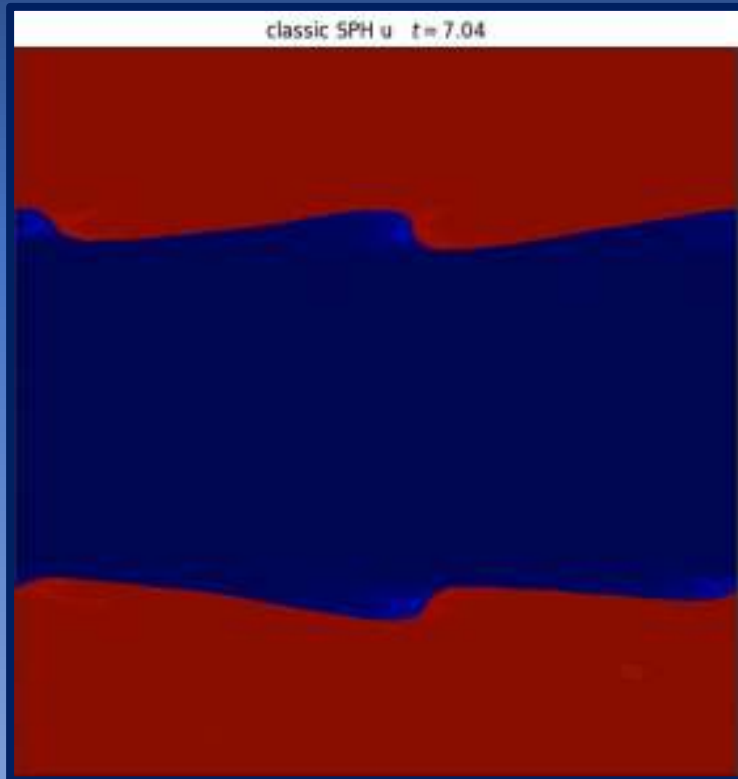
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Courtesy
M. Niemeyer,
K. Dolag



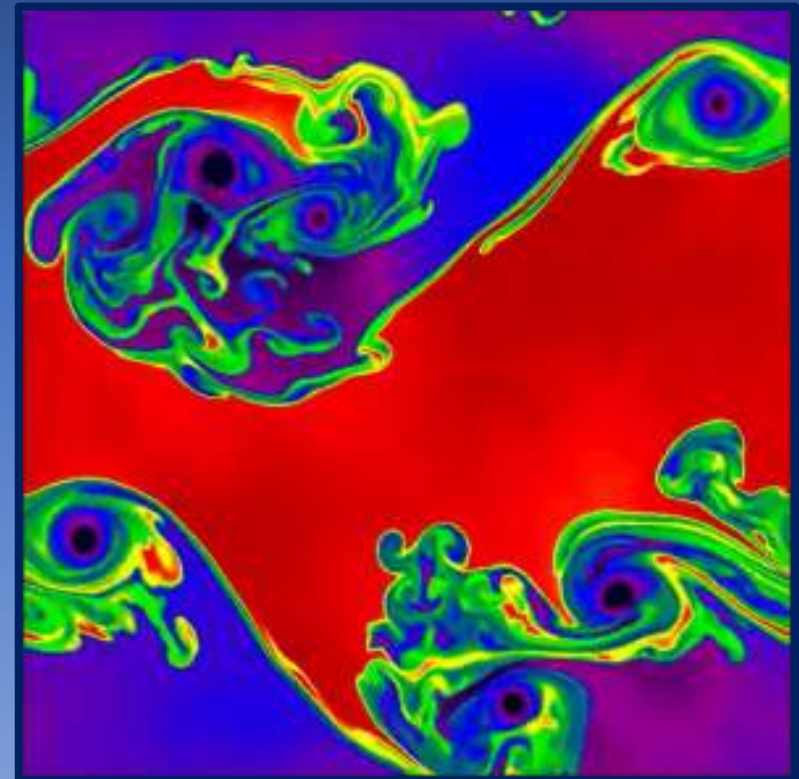
AMR

Treat gas as discretized particles:
Smoothed Particle Hydrodynamics
(Lagrangian method)



Courtesy
M. Niemeyer,
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Discretize space: Treat gas within a
grid: Adaptive Mesh Hydrodynamics
(Eulerian method)





SPH versus Grid Codes

Treat gas as discretized particles:
Smoothed Particle Hydrodynamics
(Lagrangian method)

- ✓ Very good conservation properties (mass, momentum, total energy, angular momentum, entropy)
- ✓ shape invariant
- Instabilities do not grow sufficiently
- Mixing behind shocks not sufficient

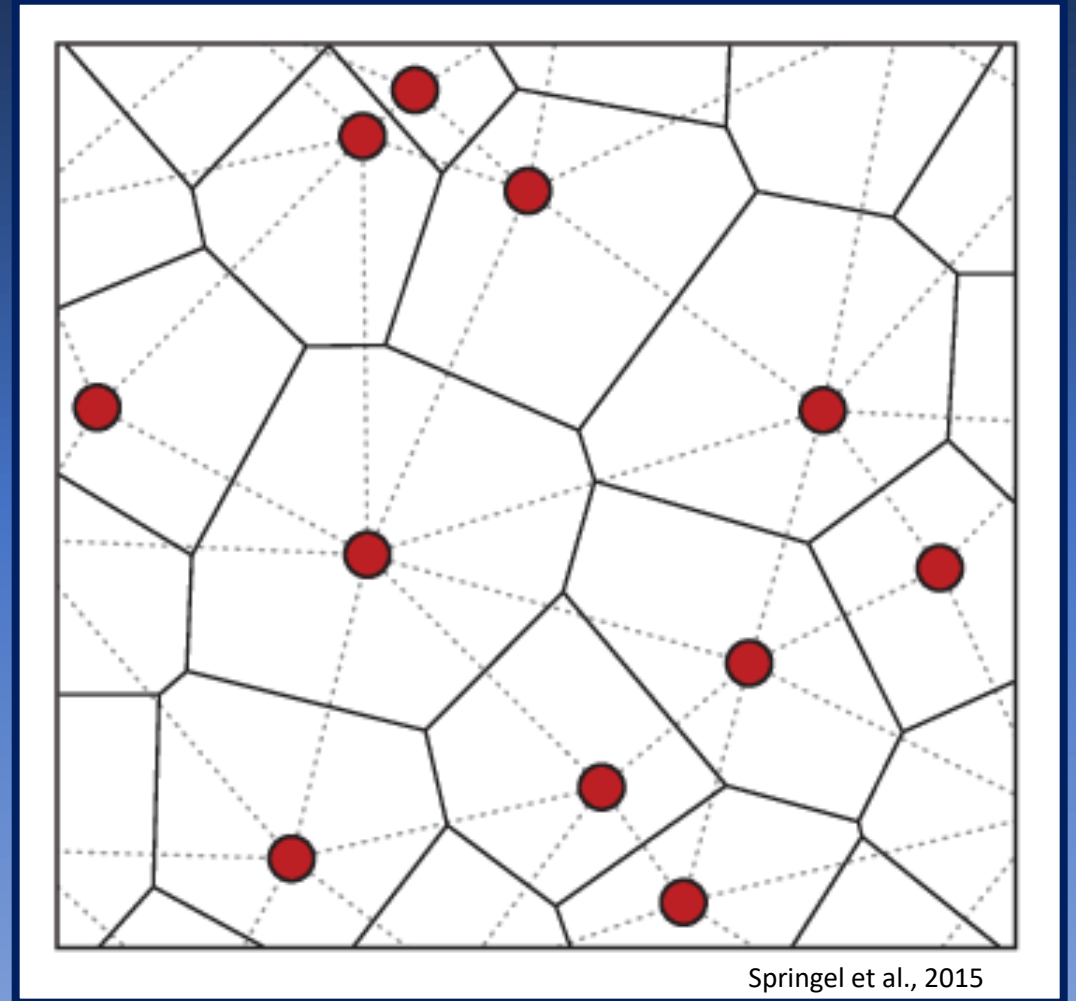
Discretize space: Treat gas within a grid: Adaptive Mesh Hydrodynamics
(Eulerian method)

- ✓ Instabilities nicely grow
- ✓ Mixing between phases works well
- Energy conservation issues (especially for fast moving elements)
- Flow over cell boundaries becomes an issue for adaptive meshes
- Not shape invariant



Merging Particle and Grid code: Moving Mesh

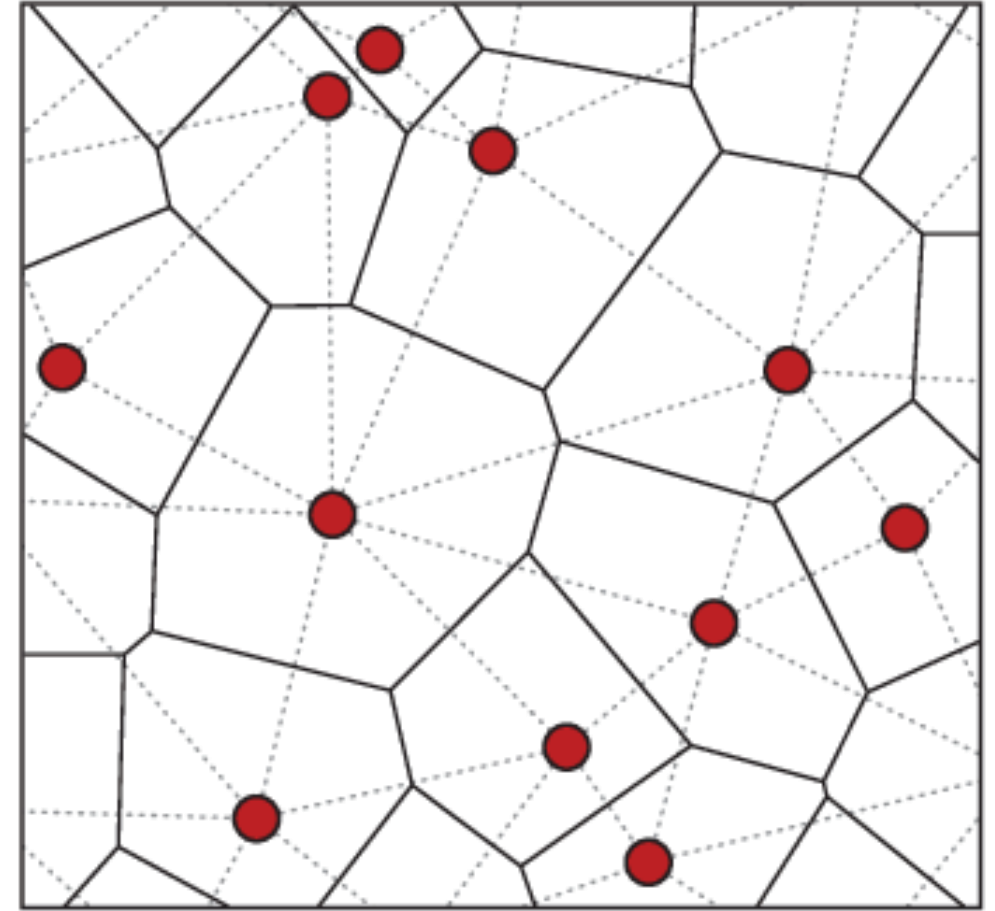
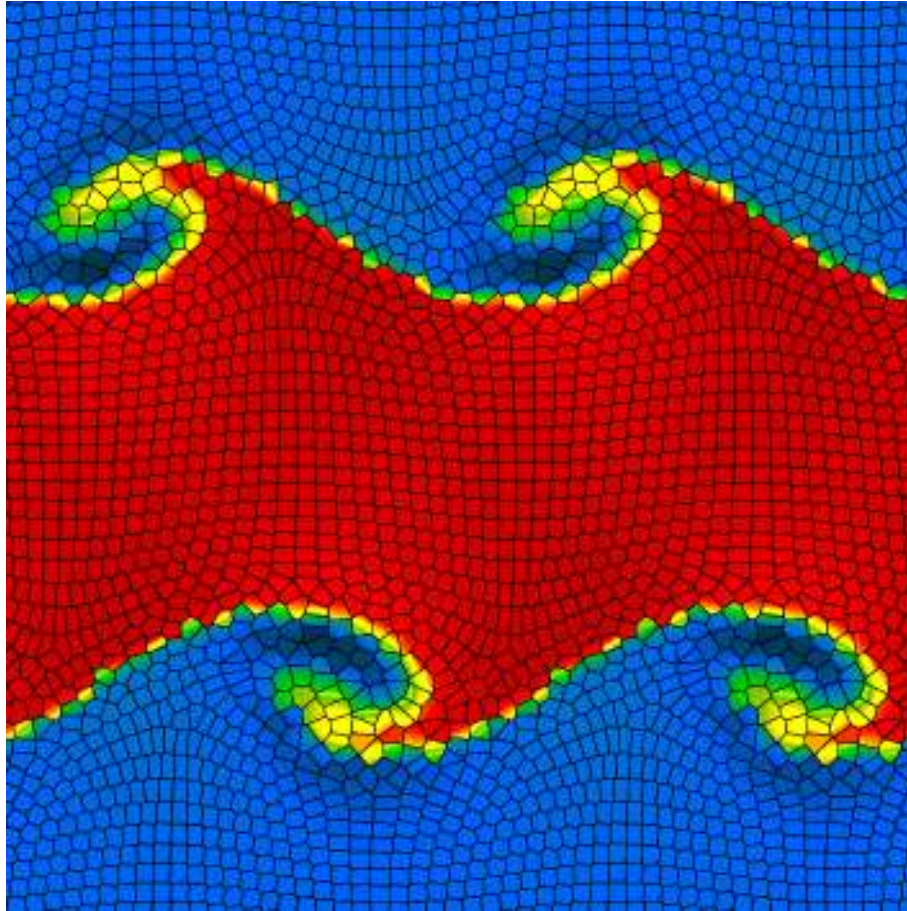
Moving Mesh Codes: Gas flows through cells, but has particle properties as well: Cells are Voronoi-tessellated





Merging Particle and Grid code

AREPO Code
by V. Springel



Springel et al., 2015

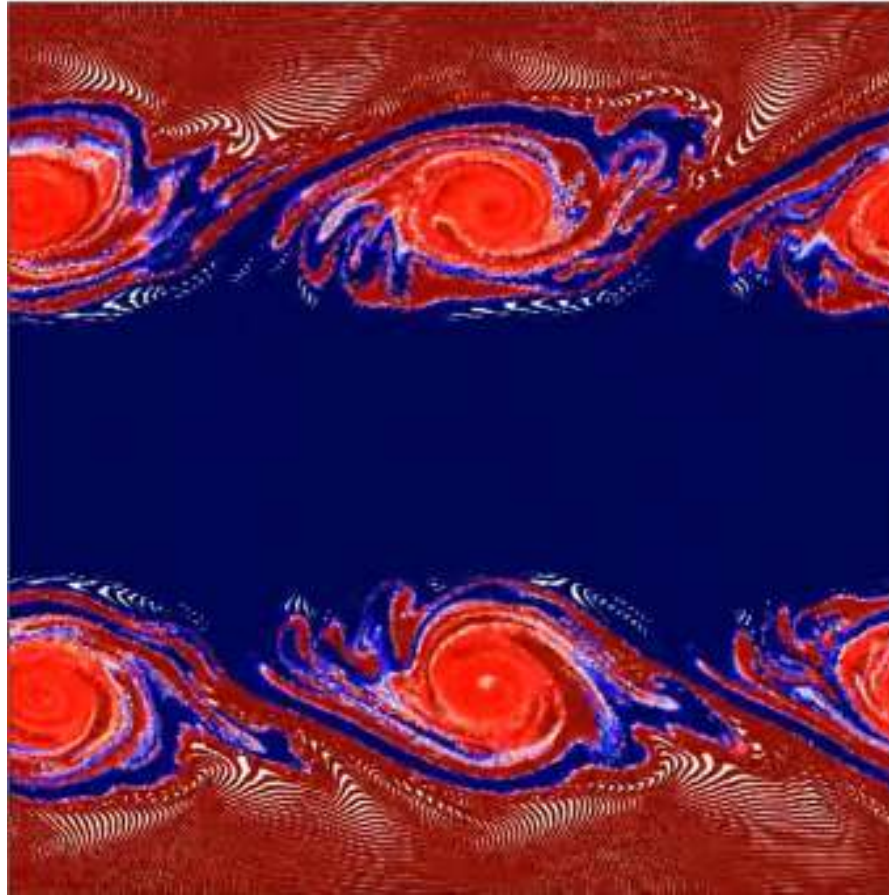


Merging Particle and Grid code

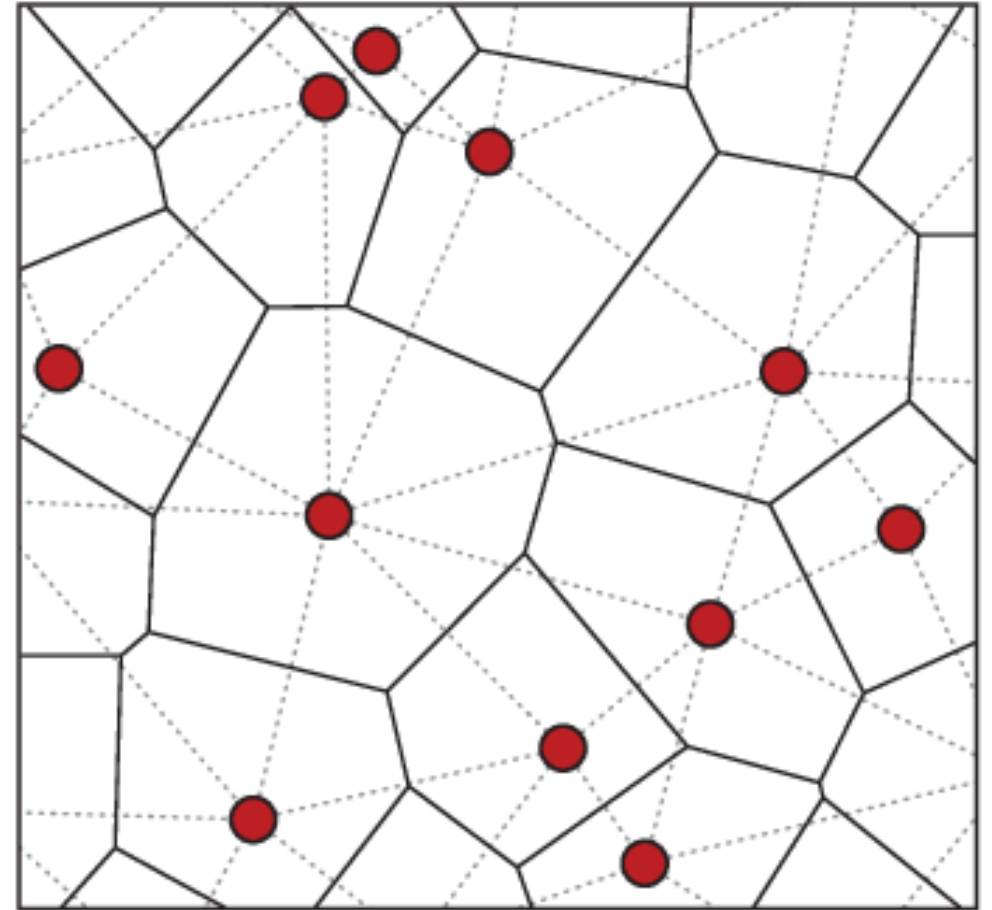
AREPO Code
by V. Springel

*

AREPO u $t = 7.34$



Courtesy M. Niemeyer, K. Dolag



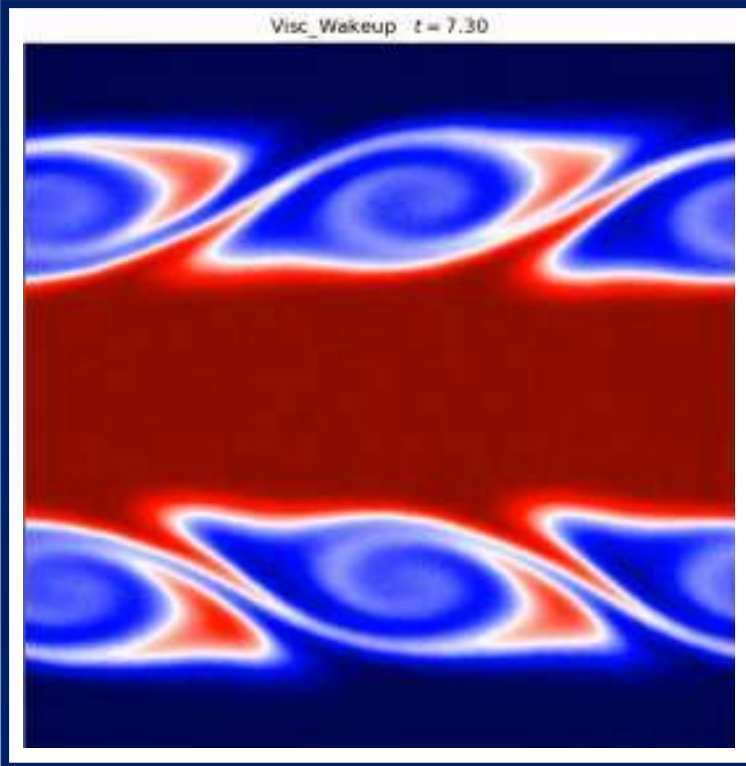
Springel et al., 2015



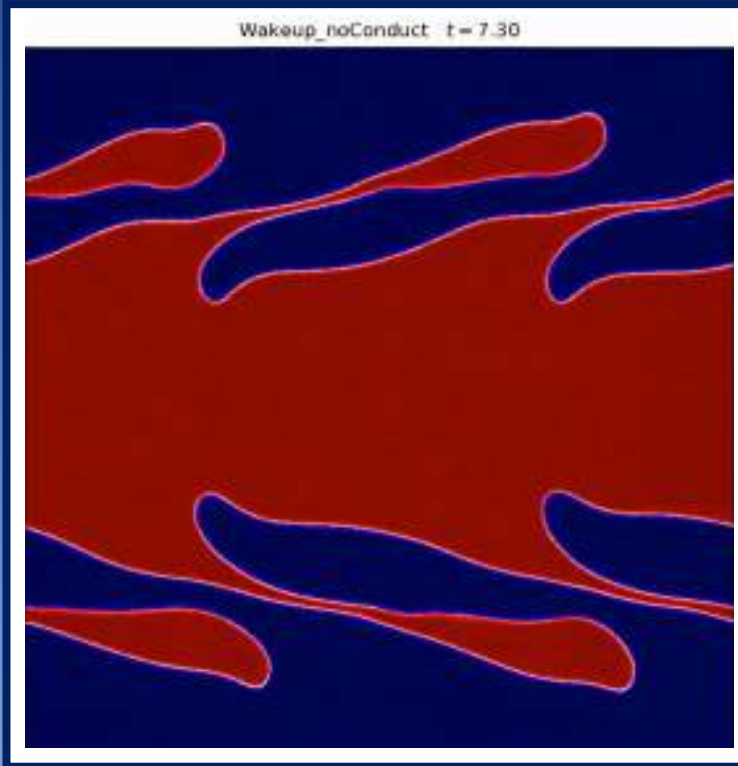
What can we do about SPH?

Introducing Artificial Viscosity and Conductivity

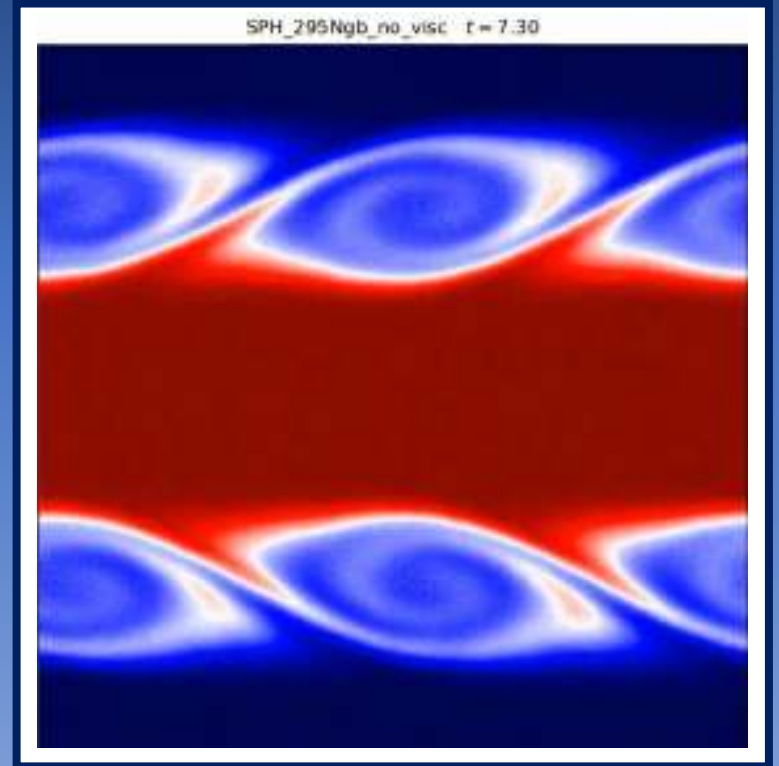
* Artificial Viscosity and Conductivity



* Artificial Viscosity only



* Conductivity only

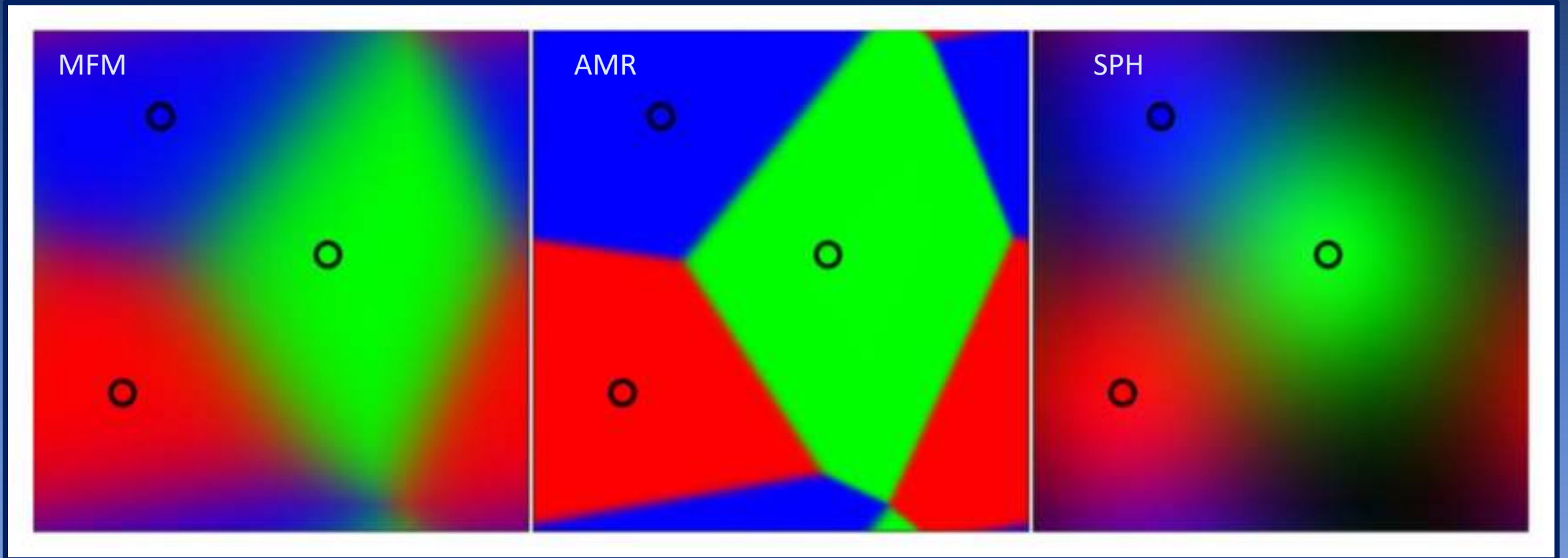


Credit: Tirso Marin



What can we do about SPH?

Meshless Finite Mass (MFM): The best of both worlds



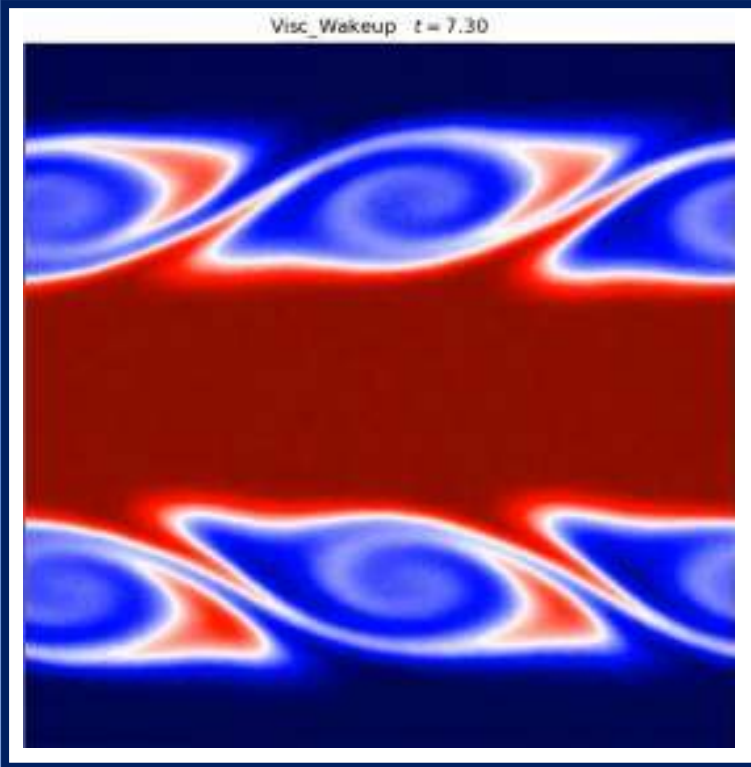
Credit: Hopkins et al 2015



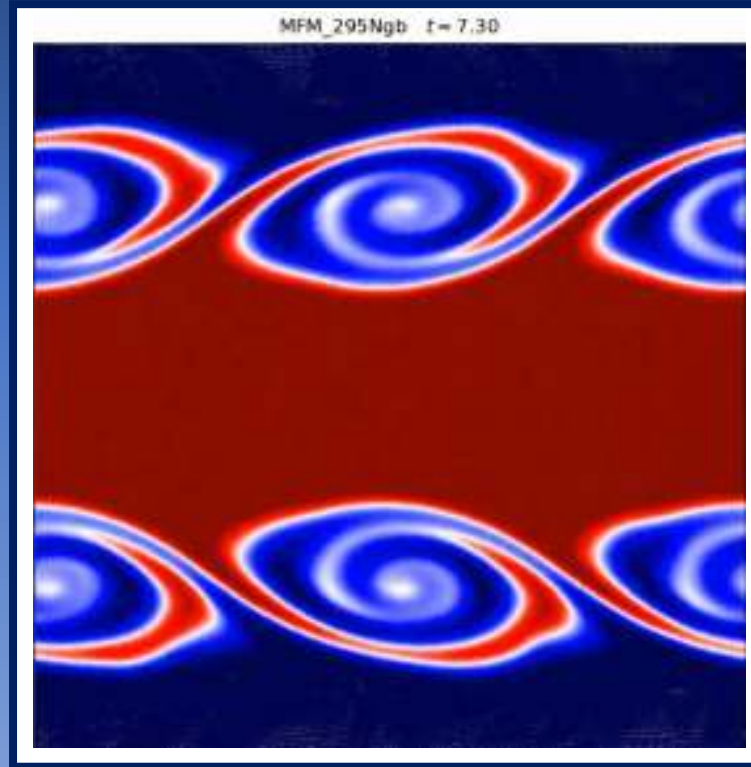
What can we do about SPH?

Meshless Finite Mass (MFM): The best of both worlds

* Artificial Viscosity and Conductivity



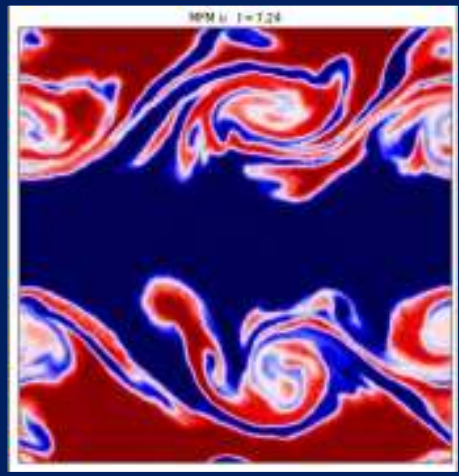
* MFM



Credit: Tirso Marin

Summary: Computational Methods

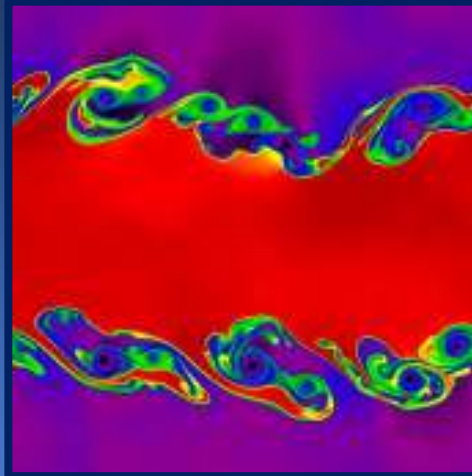
Smooth Particle Hydrodyn.



Courtesy
M. Niemeyer, K. Dolag

- ✓ Very good conservation properties (mass, momentum, total energy, angular momentum, entropy)
- ✓ shape invariant
- Instabilities do not grow sufficiently
- Mixing behind shocks not sufficient
- Shocks captured by artificial viscosity

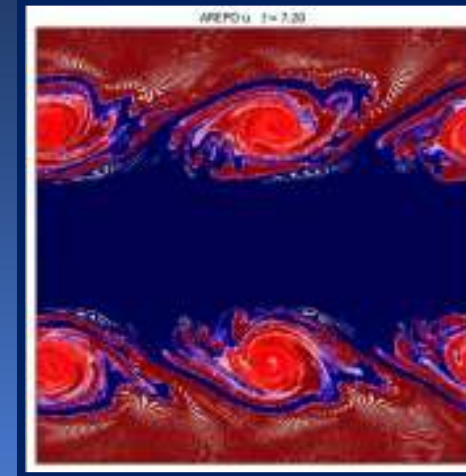
Adaptive Mesh Refinement



<https://www.astro.princeton.edu/~jstone/Athena/tests/kh/kh.html>

- ✓ Instabilities nicely grow
- ✓ Mixing between phases works well
- Energy conservation issues (especially for fast moving elements)
- Flow over cell boundaries becomes an issue for adaptive meshes
- Not shape invariant

Moving Mesh



Courtesy
M. Niemeyer, K. Dolag

- ✓ All good things from the other two
- Flow over cell boundaries (only pseudo-Lagrangian)